Software Requirements Specification

for

OpenBench

A policy-pluggable internal marketplace for people, exclusive resources, and metered capacity

Version 0.3

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Revision History

Name	Date	Reason for changes	Version
J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev	2026-09-19	Initial draft assembled from the design decisions log (D1–D17).	0.1
J. Nana	2026-09-21	Front matter and references revised; example identifier corrected.	0.2
J. Nana	2026-10-05	Overlapping-Loan rule made an eligibility rule (one Loan at a time) and Shortlist order changed, after grader feedback; inspection participation made explicit; test database changed from an in-memory database to PostgreSQL in a container to match the design.	0.3

How to read this document. Section 1 is the Vision and Scope statement. Sections 2–6 are the specification itself, and the appendices hold the glossary, the data and analysis models, the use cases, the inspection report, and the traceability table. Functional requirements carry dot-notation identifiers such as **Draft.Submit.Afford**; business rules are numbered BR-n; use cases UC-n are in Appendix D. Capitalized words such as Request, Hold, and Loan are defined terms; the glossary in Appendix A is generated from the team's shared vocabulary file.

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1 Vision and Scope

1.1 Background, Business Opportunity, and Customer Needs

A company holds scarce internal capacity of several kinds: people's time, rooms and lab instruments, GPU hours and license seats. Much of it sits idle because nobody can see it: Managers keep engineers between projects, rooms are block-booked and unused, and shared GPU pools fill with queues in which the loudest Team wins. Existing tools each cover one slice: calendars allocate by hand, talent marketplaces match only people, cluster schedulers meter only compute. Nobody can ask for "two engineers with PyTorch experience for six weeks, four GPUs for two of those weeks, and a lab room on Thursdays" in one place and get one answer. The project proposal cites internal talent marketplaces at Unilever (over 8,000 employees and 300,000 hours redeployed) and Schneider Electric (roughly 127,000 hours of unused capacity surfaced within weeks) as evidence that visibility alone recovers substantial capacity.

1.2 Business Objectives and Success Criteria

BO-1 Raise the use of shared Exclusive and Metered Resources by making free capacity visible and requestable in one place.

BO-2 Make lending a person rewarding rather than costly, so Managers release people between projects instead of holding them.

BO-3 Replace ad-hoc negotiation for capacity with one bounded, auditable process whose outcome every party can see.

SC-1 In the final demonstration one Request spanning all three Allocation Classes reaches Allocated end to end, with the Charges and Earnings visible in both Teams' Statements.

SC-2 From the first Period of use the dashboard shows utilization and idle capacity per Resource Type and credits earned and spent per Team.

SC-3 No Request waits on a silent Manager longer than the Objection Window; every Request ends in Allocated, Waiting, Refused, or Withdrawn with a complete state history.

SC-4 An untrained Manager drafts and submits a three-line Request in under five minutes, verified with at least three people from outside the team.

1.3 Vision Statement

For Managers who need people, rooms, or compute that other Teams control, OpenBench is an internal marketplace that takes one Request spanning all three kinds of capacity and allocates each part under the policy that fits it: skill matching with the lending Manager's approval for people, conflict detection with a waitlist and priority bumping for rooms and instruments, and capacity and quota checks for metered pools. Unlike a booking calendar, a talent marketplace, or a cluster scheduler, it covers all three in one Request and ties them together with a Ledger of credits in which lending a person earns what a room or a GPU costs. No money moves: credits are showback, not billing.

1.4 Major Features

Each feature is specified in the matching subsection of Section 4.

FE-1 Inventory and Resource Types as data

FE-2 Member profiles: skills and availability

FE-3 Suggestions from Members to their Manager

FE-4 Draft and submit a Request with a locked Quote

FE-5 Matched allocation: Shortlist and one approval

FE-6 Exclusive allocation: conflicts, waitlist, Bump

FE-7 Metered allocation: Pool Capacity and Quota

FE-8 Request lifecycle with bounded Holds

FE-9 Release with a partial Refund

FE-10 Ledger and pricing: Grants, Charges, Earnings, Price Bands

FE-11 Showback dashboard and Statement

FE-12 Administration and Capacity Planner settings

1.5 Scope of Initial and Subsequent Releases

Release 1 is committed for the semester. Stretch items are written as Low-priority requirements in Section 4 and built only if time allows. Future items are named so the design leaves room for them; they have no requirements here.

Area	Release 1	Stretch	Future
Request	One Request spanning all three classes; Required and Optional lines; Suggestions; dashboard and Statement	—	Unmet-demand analytics over declined Suggestions
Matched	Skill filter, availability, Shortlist, one approval by the lending Manager	—	Person consent; skill-based Day Rates
Exclusive	Conflict detection, waitlist, Priority lines that Bump	—	Priority by a Team rank set by the Capacity Planner
Metered	Pool Capacity check, per-Team Quota	Service tiers and preemption (Priority lines generalized to Metered)	—
Ledger and pricing	Grant, Charge, Earning, Refund; granted credits expire, earned carry over; Refund Schedule; Unit Price, Price Bands, priority premium; Quote locked at submission	Actual usage recorded; requested vs. used; demand surcharge	Inter-Team transfers; forecast-accuracy bonuses; usage feeds; auctions
Platform	Username and password; in-app notifications; browser only; no integrations	Email notifications	SSO; calendar, HR, and cluster-scheduler integrations

1.6 Limitations and Exclusions

LI-1 One company, one time zone, one deployment; browser only, no mobile layout; no high availability.

LI-2 No real money moves. Credits are showback; there are no invoices and no finance system.

LI-3 No external integrations in Release 1: no calendar, HR, identity, or cluster scheduler.

LI-4 A Person is notified of a Loan but cannot decline it in Release 1.

1.7 Stakeholder Profiles

Stakeholder	Major value	Attitudes, interests, constraints
Corporate management	Recovered idle capacity; less hoarding; visibility of who uses what	Wants showback numbers, not a billing bureaucracy; will not fund real chargeback
Managers	One place to ask for people, rooms, and compute; a bounded answer	Fear losing people; want Priority kept for real emergencies; little time to learn a tool
Members (as Persons)	See their own skills and Loans; can suggest needs	Want to be asked, not moved; cannot decline a Loan in Release 1
Resource Owners	Accurate Inventory; no approval chores	Want capacity changes respected by allocations; maintain their own Resources only
Capacity Planner	Levers that shape demand: Grants, prices, schedules	Wants escalations rare and settleable with a Grant; sets policy, administers no users
Administrator	Resource Types without code; simple role management	Wants no policy decisions; configures, sets no prices

1.8 Business Risks

Probability and impact are rated low, medium, or high.

RI-1 Managers do not lend, so the marketplace is empty (medium, high). Mitigation: Matched lines cost nothing, every Loan earns the Day Rate, earned credits never expire.

RI-2 Holds block rooms that end up released (medium, medium). Mitigation: no Hold outlives the Objection Window; the waitlist fills the slot.

RI-3 Badly set Grants or prices cause hoarding or idle capacity (medium, medium). Mitigation: Price Bands, mid-Period Grant adjustments, the idle-capacity view.

RI-4 Managers over-reserve and Release late (medium, medium). Mitigation: the Refund Schedule never reaches 100% and drops sharply inside seven days.

RI-5 Four people, one semester (high, high). Mitigation: Release 1 is fixed; Stretch and Future items are policy plug-ins behind the same allocator interface.

2 Introduction

2.1 Purpose

This Software Requirements Specification describes the functional and nonfunctional requirements for Release 1 of OpenBench, an internal marketplace in which a Team's Manager posts one Request for people, exclusive resources, and metered capacity and the system allocates each part under the policy that fits its Allocation Class, settled in credits. Unless a requirement is marked Stretch, it is committed for Release 1. The document is written for the development team, the course staff who grade and inspect it, and the stakeholders named in Section 1.

2.2 Project Scope and Product Features

Release 1 delivers the twelve major features FE-1 to FE-12 listed in Section 1: Resource Types as data and an Inventory; Member profiles; Suggestions; drafting and submitting a Request with a locked Quote; the three allocators (Matched, Exclusive, Metered) behind one interface; the Request lifecycle with Holds bounded by the Objection Window; Release with a partial Refund; the Ledger with Grants, Charges, Earnings, Refunds, and Price Bands; a showback dashboard and Statement; and administration. The Vision and Scope section states what is deferred to Stretch and Future and why.

2.3 Document Conventions

- **Shall** marks a mandatory requirement. Priorities are *High* (Release 1 is not usable without it), *Medium* (Release 1 is complete with it), and *Low* (Stretch).
- Functional requirements carry a dot-notation identifier whose first part names the feature, for example **Draft.Submit.Afford**. Business rules are BR-n, use cases UC-n, features FE-n, and the other identifier prefixes are introduced where they appear.
- Capitalized terms such as Request, Hold, Loan, and Objection Window are defined in the glossary, Appendix A. The document uses the everyday word a Manager would use and avoids money words: a Credit is the unit of account and never a direction, and nothing is paid, funded, or billed.
- Default values marked *example default* are settings of the Capacity Planner; the value shown is the one the team will ship and test with, and it can be changed as data.

2.4 References

1. J. Nana, B. Lujan, F. Turopov, *OpenBench: A Policy-Pluggable Internal Resource Marketplace*, CSDS 393 informal project proposal, 2026.
2. CSDS 393, *Rubric for Software Requirements Specification*, 2026.
3. Project vocabulary file `CONTEXT.md` and decisions log, project repository, 2026.

2.5 Definitions

All defined terms are collected in Appendix A, which is generated from the project's vocabulary file at build time. The terms a first-time reader needs are: a *Team* has one *Manager* and one *Account* of credits; a *Request* is a set of *Request Lines*, each of exactly one *Allocation Class* (*Matched* for people, *Exclusive* for things held whole over a window, *Metered* for divisible pools); a satisfied Matched line is a *Loan*; a *Hold* is a tentative, uncharged allocation that lives no longer than the *Objection Window*; and the *Ledger* records four kinds of *Transaction*: *Grant*, *Charge*, *Earning*, and *Refund*.

3 Overall Description

3.1 Product Perspective

OpenBench is a new, self-contained web application. It replaces the mix of booking calendars, spreadsheets, chat messages, and hallway negotiation by which Teams currently obtain people, rooms, and compute from one another. It has no external system interfaces in Release 1 (LI-3); every flow in Figure 1 is between the system and one of its five actors. The system consists of a

single server process that serves a browser front end and a REST API, backed by one relational database that holds the Inventory, the Requests, and the Ledger.

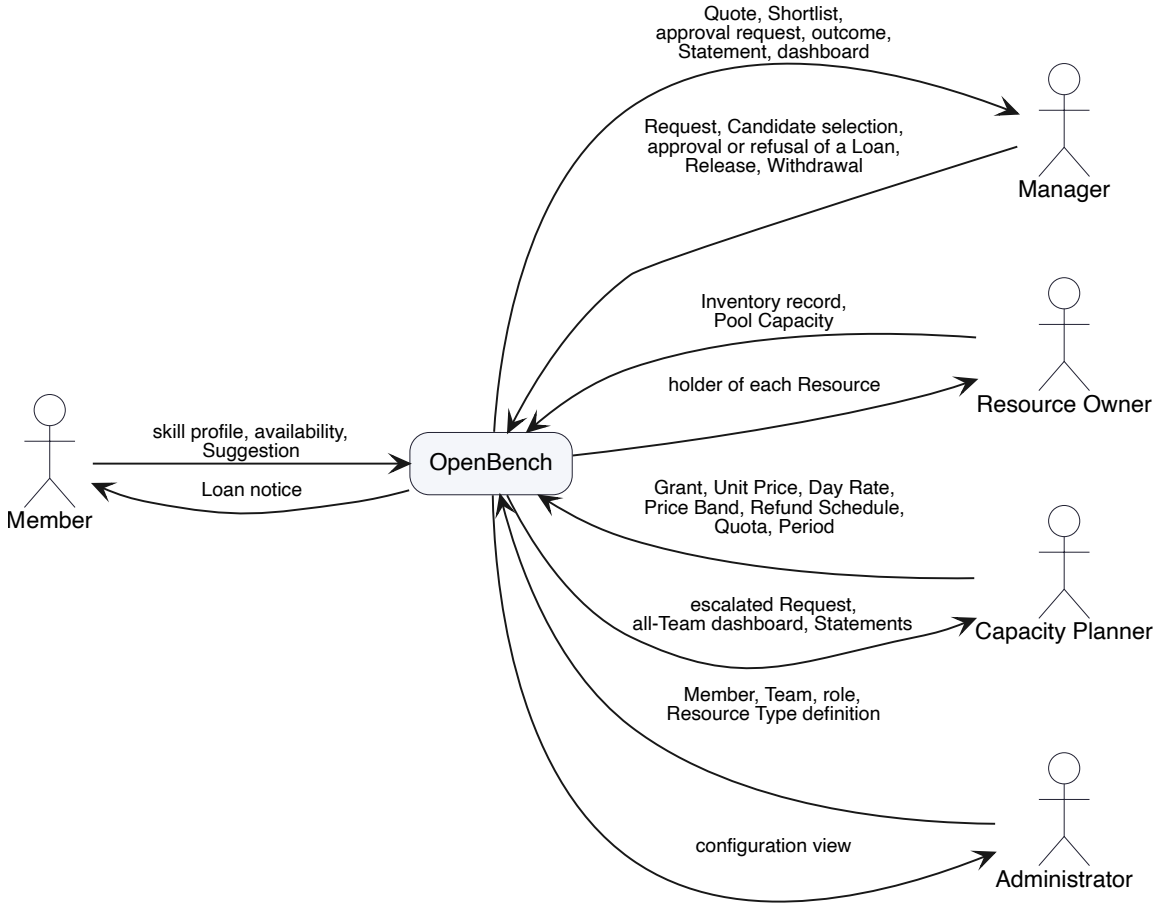


Figure 1: Context diagram. The five actors are roles held by Members; one Member may hold several.

3.2 User Classes and Characteristics

Every user is a *Member*. Four further roles are additive and are granted by the Administrator. One Member may hold several roles, for example a Manager who is also the Capacity Planner. Roles are enforced on the server for every action (SE-1).

User class	Characteristics and what they do in the system	Importance
Member	About 500 people, any level of technical skill, using the system a few times a month. Maintains their own skills and availability, makes Suggestions to their own Manager, sees their Loans. Lendable as a Person.	Favored (as Person)
Manager	About 50 people, one per Team. The only actor who drafts and submits Requests, selects Candidates, approves or refuses Loans of their own Team's Members, Releases lines, and reads the Team's Statement. Uses the system weekly.	Most important
Resource Owner	A few dozen people responsible for the Inventory records of specific Exclusive and Metered Resources, including Pool Capacity. Never approves an allocation.	Important
Capacity Planner	One or two people. Sets each Team's Grant per Period, Unit Prices, Price Bands, the Day Rate, Refund Schedules, Quotas, the priority premium, the bump cutoff, the Objection Window, and the Period. Receives escalations and settles them, often with a mid-Period Grant adjustment. Sees every Team's dashboard and Statement.	Most important
Administrator	One or two IT staff. Creates Members and Teams, grants roles, maintains the skill list, and defines Resource Types as data. Configures the system and sets no policy.	Important

3.3 Operating Environment

- **Server:** one on-premises server running one Java 17 process (Spring Boot) that serves both the REST API and the built front end as static files; delivered as a Docker image. An nginx reverse proxy is used only when TLS termination is wanted in front of the process.
- **Database:** PostgreSQL 16; automated tests run against PostgreSQL 16 in a container started by the test suite (Testcontainers), with the same schema migrations as production.
- **Clients:** current versions of Chrome, Firefox, Safari, and Edge on desktop; no mobile layout.
- **Organization:** one company, one time zone; all windows, Price Bands, and Periods are expressed in that time zone.

3.4 Design and Implementation Constraints

CO-1 The back end shall be written in Java 17 with Spring Boot and built with Maven; the front end shall be written in TypeScript with React and Vite and built into the Spring Boot artifact by the same Maven build.

CO-2 Persistent data shall be stored in PostgreSQL 16; automated tests shall run against PostgreSQL 16 in a container started by the test suite (Testcontainers) with the same schema migrations as production.

CO-3 The front end shall talk to the back end only through the REST/JSON API, and every action available in the user interface shall be available through the API with the same role checks.

CO-4 Commitment of Holds and posting of the corresponding Charges and Earnings shall be executed inside one database transaction (BR-7).

CO-5 Time-driven behavior (Objection Window lapse, Hold and waitlist expiry, Period rollover)

shall be implemented as scheduled jobs inside the server process, not as external cron jobs.

CO-6 Authentication shall be by username and password with passwords stored hashed; roles shall be enforced with Spring Security on the server.

CO-7 Each Allocation Class shall be implemented as one allocator behind a common interface, so that a new class is added by implementing that interface without modifying the existing allocators (QA-7).

3.5 User Documentation

UD-1 A short user guide, one page per role, delivered as pages inside the application: how to draft a Request, how to read a Quote and the Available Balance, how to approve a Loan, how to Release a line, how to read a Statement.

UD-2 An installation and configuration guide covering the Docker image, database setup, the first Administrator account, and the Capacity Planner settings with their example defaults.

UD-3 Field-level help text for every setting of the Capacity Planner and every column of the Draft Request screen.

3.6 Assumptions and Dependencies

AS-1 A Capacity Planner exists and sets a Grant for every Team before the first Period starts; without Grants no Exclusive or Metered line is affordable.

AS-2 Resource Owners keep the Inventory current, in particular Pool Capacity; the system detects conflicts against its own records only.

AS-3 Managers record Loans truthfully; the system has no way to verify that a lent Person actually worked for the borrowing Team.

AS-4 Availability is what a Member records in their profile; no calendar is consulted.

DE-1 The system depends on Java 17, Spring Boot, PostgreSQL, and current desktop browsers being available on the company's infrastructure.

DE-2 The project's build and documentation toolchain is pinned in the repository's Nix flake; teammates need Nix to reproduce the build.

4 System Features

4.1 Inventory and Resource Types

4.1.1 Description and Priority

The Inventory is the record of every Resource that exists, its Resource Type, and who currently holds it or has it on Hold. A Resource Type is data, created by an Administrator, that fixes a Resource's Allocation Class and attributes; adding a new kind of room, instrument, or pool therefore needs no

code. Resource Owners maintain the records of Exclusive and Metered Resources, including Pool Capacity. This feature realizes FE-1 and is exercised by UC-1 and UC-2. **Priority:** High.

4.1.2 Stimulus/Response Sequences

- **Stimulus:** An Administrator defines a Resource Type. **Response:** The system stores it and Request Lines can name it immediately.
- **Stimulus:** A Resource Owner adds a Resource or changes the units of a Metered Resource. **Response:** The system updates the Inventory, refusing a reduction that would fall below what is already allocated or held.
- **Stimulus:** A Member with any role views a Resource for a time window. **Response:** The system shows whether it is free, on Hold, or allocated, and to which Team.

4.1.3 Functional Requirements

Inventory.Type.Define The system shall let an Administrator define a Resource Type as data consisting of a unique name, exactly one Allocation Class (Matched, Exclusive, or Metered), and zero or more named attributes, each of type text, number, date, or a list of allowed values.

Inventory.Type.NoCode Defining, changing the attributes of, or retiring a Resource Type shall require no change to program code or deployment configuration.

Inventory.Type.Person The system shall provide one built-in Resource Type named Person, of class Matched, whose Resources are the Members; it shall not be retired.

Inventory.Type.Retire The system shall let an Administrator retire a Resource Type. A retired type shall accept no new Request Lines; existing allocations and Transactions that reference it shall be unaffected. A Resource Type shall never be deleted once any Request Line references it.

Inventory.Resource.Add The system shall let a Resource Owner add an Exclusive or Metered Resource by giving it a unique name within its Resource Type, choosing the type, and filling in the type's attributes. The Resource Owner who adds a Resource becomes its owner and may transfer ownership to another Resource Owner.

Inventory.Resource.Retire The system shall let a Resource Owner retire a Resource from a date onward. The system shall refuse the retirement while any allocation or Hold on that Resource lies on or after that date, and shall name the blocking Request Lines.

Inventory.Pool.Units A Metered Resource shall record the number of units it supplies at once. The Pool Capacity of a Metered Resource Type shall be the sum of the units of its Resources that are not retired.

Inventory.Pool.Reduce When a Resource Owner reduces the units of a Metered Resource, the system shall refuse the change if, in any future hour, the units already allocated or held for that Resource Type would exceed the new Pool Capacity, and shall report the earliest such hour and the shortfall.

Inventory.View.Holder The system shall show, for any Resource and any time window, whether it is free, on Hold, or allocated in each part of the window, and for Holds and allocations the

holding Team and Request.

Inventory.View.Filter The system shall let a Member list Resources filtered by Resource Type, by attribute value, and by being free for a given window.

Inventory.Owner.NoApprove The system shall place no approval step for a Resource Owner in the allocation of any line (BR-32).

4.2 Member Profiles

4.2.1 Description and Priority

Every Member has a profile: their Team, a list of skills, and their availability. Profiles are the data against which Matched lines are evaluated. Members maintain their own skills and availability; the Administrator sets Team membership. This feature realizes FE-2. **Priority:** High.

4.2.2 Stimulus/Response Sequences

- **Stimulus:** A Member edits their skills or availability. **Response:** The system saves the profile and, for any Waiting Matched line the Member now qualifies for, rebuilds that line's Shortlist (Matched.Shortlist.Rebuild).
- **Stimulus:** A Manager opens their Team's roster. **Response:** The system shows each Member's skills, availability, and current and upcoming Loans.

4.2.3 Functional Requirements

Profile.Team Every Member shall belong to exactly one Team, set by an Administrator, and the profile shall show it.

Profile.Skills A Member shall maintain their own list of skills, each chosen from the skill list maintained by an Administrator (Admin.Skill.List), so that skill matching is exact.

Profile.Availability A Member shall record their availability as a set of date ranges during which they may be lent. A Member is available for a window when every day of the window lies inside a recorded range. Forming, shortening, or ending a Loan shall not change these ranges.

Profile.Availability.Loans A Person shall be in at most one Loan at a time: a Member with a Loan overlapping any day of a Matched line's window shall not be a Candidate for that line, even when available for the whole window (Matched.Shortlist.Build, BR-13). A Loan that ends early (Release.Loan.End) shall free the Person from the day after its new last day.

Profile.View.Own A Member shall see their own profile and their current and upcoming Loans, with the borrowing Team and window of each.

Profile.View.Manager A Manager shall see the profiles, availability, and Loans of every Member of their own Team, and shall be able to edit a Member's availability on their behalf.

Profile.Data.Limit A profile shall hold no personal data other than name, email address, Team, skills, and availability (SE-4).

4.3 Suggestions

4.3.1 Description and Priority

A Suggestion is a single Request Line that a Member proposes to their own Manager: “we need four GPUs in weeks 42–43” or “bring on Dana for the pilot.” It has no allocation effect until the Manager accepts it into a Request. Several Suggestions may be accepted into the same line, which is how three engineers asking for the same thing are grouped. Each Suggestion keeps its original ask, so what was asked can later be compared with what was formalized; declined Suggestions are kept as the raw data for the Future unmet-demand analytics. This feature realizes FE-3 and is exercised by UC-3. **Priority:** Medium.

4.3.2 Stimulus/Response Sequences

- **Stimulus:** A Member submits a Suggestion. **Response:** The system stores it as Open and shows it in the Manager’s Open list.
- **Stimulus:** The Manager accepts a Suggestion into a Draft Request. **Response:** The system creates or extends a line, marks the Suggestion Accepted, links it to the line, and tells the Member.
- **Stimulus:** The Manager declines a Suggestion. **Response:** The system marks it Declined and tells the Member, with the reason if one was given.

4.3.3 Functional Requirements

Suggest.Create The system shall let a Member create a Suggestion addressed to their own Manager, containing exactly one proposed Request Line: a Resource Type, a quantity, a time window, for a Matched line either a skill set or a named Person, and an optional note.

Suggest.States A Suggestion shall be in exactly one of the states Open, Accepted, or Declined. Only Open Suggestions may change state.

Suggest.Accept.New The system shall let the Manager accept an Open Suggestion into a Draft Request of their Team as a new Request Line pre-filled from the Suggestion.

Suggest.Accept.Existing The system shall let the Manager accept an Open Suggestion into an existing line of a Draft Request that has the same Resource Type, so that several Suggestions point at one line.

Suggest.Link An Accepted Suggestion shall record the Request Line it became. The link shall survive later edits to the line.

Suggest.Keep The system shall keep a Suggestion’s original ask unchanged after acceptance, even if the Manager edits the line, so that asked and formalized can be compared.

Suggest.Decline The system shall let the Manager decline an Open Suggestion with an optional reason, and shall notify the Member.

Suggest.NoEffect A Suggestion shall have no effect on the Inventory, Holds, or the Ledger until the line it was accepted into is submitted as part of a Request.

Suggest.View A Member shall see the state of each of their Suggestions; a Manager shall see all Suggestions addressed to them, Open ones first.

Suggest.Retain The system shall never delete a Suggestion; Declined and Open Suggestions are retained for the data-retention period (QA-8).

4.4 Draft and Submit a Request

4.4.1 Description and Priority

A Manager assembles a Request in the Draft state, by hand or by accepting Suggestions, marks each Request Line Required or Optional, sees a Quote per line, and submits. Submission is refused when the Required lines together cost more than the Team's Available Balance. Quotes are locked at submission: the Charge that later posts equals the Quote. This feature realizes FE-4 and is exercised by UC-4. **Priority:** High.

4.4.2 Stimulus/Response Sequences

- **Stimulus:** The Manager adds or edits a line. **Response:** The system shows the line's Quote at current prices and the totals for Required and Optional lines against the Available Balance.
- **Stimulus:** The Manager submits. **Response:** If the Required lines are affordable the system locks the Quotes and evaluates the Request (Section 4.8); otherwise it shows the shortfall and leaves the Request in Draft.
- **Stimulus:** The Manager withdraws a Request. **Response:** The system releases any Holds and held credits and marks the Request Withdrawn.

4.4.3 Functional Requirements

Draft.Create The system shall let a Manager create a Request in the Draft state on behalf of their own Team, and only a Manager shall create or submit a Request (BR-1).

Draft.Line.Add The system shall let the Manager add a Request Line by choosing a Resource Type, a quantity, and a time window; the line's Allocation Class shall be taken from the Resource Type and shall not be editable.

Draft.Line.Matched For a Matched line the Manager shall give either a set of one or more skills and a quantity, or a named Person, in which case the quantity shall be one (BR-5).

Draft.Line.Exclusive For an Exclusive line the Manager shall give the Resource Type, optionally a specific Resource of that type, a quantity of instances (default one), a window, optionally a weekly recurrence (weekday, time of day, first and last week), and optionally the Priority mark.

Draft.Line.Metered For a Metered line the Manager shall give the Resource Type, a number of units to hold concurrently, and a window.

Draft.Line.Required Every line shall be marked Required or Optional; the default is Required.

Draft.Line.Split The system shall let the Manager split a line into two lines of the same Resource Type with smaller quantities or shorter windows, so that a part can be marked Optional. There shall be no per-line partial-acceptance setting (BR-3).

Draft.Line.Validate The system shall refuse to save a line whose quantity is below one, whose window does not start after the current time, whose recurrence has no occurrence inside its window, or whose Resource Type is retired.

Draft.Quote.Line While drafting, the system shall show each line's Quote computed by BR-22 at the current settings; Matched lines shall show no Quote because they cost nothing (BR-10).

Draft.Quote.Totals The system shall show the sum of Quotes of Required lines, the sum for Optional lines, and the Team's Available Balance, and shall highlight the shortfall when the Required sum exceeds the Available Balance.

Draft.Quote.Refresh When the Capacity Planner changes a price setting, Quotes of Draft lines shall be recomputed the next time the Draft is shown; Quotes of submitted lines shall not change (BR-24).

Draft.Submit.Afford The system shall refuse submission while the sum of Quotes of Required lines exceeds the Team's Available Balance, and shall show the shortfall (BR-28).

Draft.Submit.Lock On submission the system shall lock every line's Quote; the Charge posted for that line shall equal the locked Quote.

Draft.Submit.Evaluate On submission the system shall record the submission time (used for waitlist order) and evaluate every line as specified in Section 4.8.

Draft.Withdraw The system shall let the Manager withdraw a Request in the Draft, Pending, or Waiting state; all Holds and held credits release and the Request becomes Withdrawn. An Allocated Request is not withdrawn; its lines are Released (Section 4.9).

Draft.Edit.Waiting The system shall let the Manager edit a Waiting Request, which returns it to Draft: its lines leave their waitlists, held credits release, and Quotes are recomputed on resubmission.

4.5 Matched Allocation

4.5.1 Description and Priority

A Matched line asks for people. It is either a skill set plus a quantity or a named Person. For a skill set the system builds a Shortlist of Candidates who hold every listed skill, are available for the whole window, and have no Loan overlapping it (a Person is in at most one Loan at a time), ordered by fewest person-days lent in the current Period, and the requesting Manager picks; there is no automatic assignment. Each selected Candidate needs one approval from their own Manager. If the Objection Window lapses with approvals outstanding, the Request escalates to the Capacity Planner. A satisfied Matched line is a Loan; it costs the borrowing Team nothing and earns the lending Team the Day Rate. This feature realizes FE-5 and is exercised by UC-5, UC-6, and UC-7. **Priority:** High.

4.5.2 Stimulus/Response Sequences

- **Stimulus:** A Request with a skill-based Matched line enters Pending. **Response:** The system builds the Shortlist and asks the requesting Manager to select Candidates.

- **Stimulus:** The requesting Manager selects Candidates. **Response:** The system sends one approval request per Candidate to that Candidate's Manager and notifies each Candidate.
- **Stimulus:** A lending Manager approves or refuses. **Response:** On approval the line counts as satisfied for that Candidate; on refusal the requesting Manager is asked to pick a replacement and every Hold on the Request releases.
- **Stimulus:** The Objection Window lapses with an approval outstanding. **Response:** The system escalates the outstanding approvals to the Capacity Planner.

4.5.3 Functional Requirements

Matched.Line.Skill A skill-based Matched line shall be satisfied only by Candidates who hold every skill listed on the line and are available for the whole window (BR-13).

Matched.Line.Named A Matched line naming a Person shall have quantity one; the system shall check only that the Person is available for the whole window and has no Loan overlapping any day of it, and shall then request approval from that Person's Manager (BR-5).

Matched.Line.OtherTeam Candidates shall be drawn only from Teams other than the requesting Team; work inside a Team is not a Loan (BR-12).

Matched.Shortlist.Build When a Request enters Pending, the system shall build a Shortlist for each skill-based Matched line containing every Member who satisfies Matched.Line.Skill and Matched.Line.OtherTeam and has no Loan overlapping any day of the line's window; a Member with such a Loan shall not be a Candidate for the line (BR-13).

Matched.Shortlist.Order The Shortlist shall be ordered by the Candidate's person-days lent in the current Period, fewest first, counting the person-days (Ledger.Earning.Post) of every formed Loan of the Candidate that fall inside the Period, a Loan ended early only up to its new last day; Candidates with equal counts shall be ordered by name.

Matched.Shortlist.Show The Shortlist shall show each Candidate's name, Team, skills, and person-days lent in the current Period, and shall never show a Candidate's other Requests or Statements.

Matched.Shortlist.Rebuild While a Matched line is unsatisfied, the system shall rebuild its Shortlist whenever a Member's skills, availability, Team, or Loans change, and shall notify the requesting Manager when a Candidate becomes available for a line that had too few.

Matched.Select The requesting Manager shall select exactly the line's quantity of Candidates from the Shortlist. The system shall never select a Candidate automatically.

Matched.Select.Short If a Shortlist holds fewer Candidates than the line's quantity, the line shall be unsatisfied: a Required line places the Request in Waiting (Section 4.8); an Optional line is dropped when the Request is allocated.

Matched.Approve.Route For each selected Candidate the system shall send one approval request to the Manager of the Candidate's Team, stating the requesting Team, the window, and the Earning the lending Team will receive.

Matched.Approve.Decide The lending Manager shall approve or refuse each approval request individually, with an optional reason. The decision shall be recorded with actor and timestamp.

Matched.Approve.Own The system shall not send an approval request to, or accept an approval from, the requesting Manager for a Member of their own Team (BR-12).

Matched.Refuse.Replace When an approval is refused, the system shall ask the requesting Manager to select a replacement from the Shortlist. The Objection Window shall not restart (BR-8).

Matched.Refuse.None When an approval is refused and the Shortlist holds no further Candidate, or the line named a Person, the line cannot be satisfied: a Required line places the Request in Refused; an Optional line is dropped.

Matched.Escalate When the Objection Window lapses with an approval outstanding, the system shall present the outstanding approvals to the Capacity Planner, who shall approve or refuse each on the silent Manager's behalf; the silent Manager shall be notified of the outcome.

Matched.Escalate.Later After the Objection Window has lapsed, every further approval request on that Request shall go directly to the Capacity Planner.

Matched.Person.Notify The system shall notify a Candidate when they are selected and a Person when their Loan forms. The Person shall have no action to take and cannot decline (BR-14).

Matched.Loan.Form When the Request becomes Allocated, each approved Candidate on a Matched line shall become a Loan for the line's window, and an Earning of the Day Rate times the person-days of the Loan shall post to the lending Team (BR-11).

Matched.Loan.Optional An Optional Matched line whose approvals arrive after the Request is Allocated shall form its Loan and post its Earning at that time.

Matched.Loan.Visible A Loan shall appear on the Person's profile, on both Managers' Team views, and on the borrowing Team's Request.

4.6 Exclusive Allocation

4.6.1 Description and Priority

An Exclusive line asks for Resources that are held whole over a time window: a room, an instrument, a dedicated node. The system detects conflicts against existing allocations and Holds, picks a free instance when none is named, and requires every occurrence of a recurring line to be free. A Required line that cannot be satisfied joins the waitlist for its Resource Type and window; when a slot frees, the waitlist fills it automatically. A Manager may mark a line as a Priority line, at a premium on the Unit Price; a Priority line can Bump a standard line outside the bump cutoff and cannot itself be Bumped. This feature realizes FE-6 and is exercised by UC-4, UC-8, and UC-10. **Priority:** High; the Priority and Bump requirements are Medium.

4.6.2 Stimulus/Response Sequences

- **Stimulus:** A Request with an Exclusive line is evaluated. **Response:** If a free instance exists for the whole window the line is held or allocated; otherwise a Required line joins the waitlist and the Manager sees its position.

- **Stimulus:** A Priority line finds no free instance. **Response:** Outside the bump cutoff the system Bumps the standard line that blocks it, refunds that line in full, moves it to the head of the waitlist, and notifies its Manager.
- **Stimulus:** An allocation or Hold ends early. **Response:** The system offers the slot to the waitlist in order (Section 4.8).

4.6.3 Functional Requirements

Exclusive.Line.Form An Exclusive line shall name a Resource Type and optionally one specific Resource of that type, a quantity of instances, a window, and optionally a weekly recurrence.

Exclusive.Conflict An instance shall be free for a window when no allocation and no Hold on that instance overlaps any part of the window; a line shall be satisfiable when its quantity of free instances exists for every occurrence (BR-3).

Exclusive.Recurrence For a recurring line every occurrence shall be free on the same instance; if any occurrence is not free the whole line is unsatisfiable (BR-3). To accept fewer occurrences the Manager splits the line (Draft.Line.Split).

Exclusive.Pick When no specific Resource is named the system shall choose, among the free instances, the one with the fewest allocated hours in the current Period, and shall record the choice on the line.

Exclusive.Waitlist.Join A Required Exclusive line that is unsatisfiable shall join the waitlist for its Resource Type and window; an Optional line shall never join a waitlist (BR-18).

Exclusive.Waitlist.Order Each waitlist shall be ordered with Priority lines first, then by the submission time of the Request; a Bumped line shall be placed at the head (BR-18).

Exclusive.Waitlist.Fill When capacity frees through a Release, a Bump, an expired Hold, or a retirement being undone, the system shall offer it to the waiting lines in order and satisfy the first line whose window fits, as specified by Lifecycle.Waiting.Reevaluate; the Manager of that line shall be notified.

Exclusive.Waitlist.Expire A waiting line shall leave the waitlist when its window start passes; its Request shall stay Waiting until the Manager edits or withdraws it.

Exclusive.Waitlist.View The Manager shall see each waiting line's position and the number of lines ahead of it.

Exclusive.Priority.Mark The system shall let the Manager mark an Exclusive line as a Priority line while drafting; its Quote shall include the priority premium (BR-15). Priority lines shall be offered for Exclusive lines only in Release 1 (BR-19).

Exclusive.Priority.Bump When a Priority line finds no free instance, the system shall Bump the standard lines, allocated or on Hold, that block the instance needing the fewest Bumps, preferring among equal counts the lines with the latest submission time, provided every Bump satisfies Exclusive.Priority.Cutoff.

Exclusive.Priority.Cutoff The system shall not Bump a line whose window, or any occurrence of it, starts within the bump cutoff of the Resource Type from the current time (example

default 24 hours; BR-16). A Priority line that cannot Bump shall join the waitlist ahead of all standard lines.

Exclusive.Priority.NoBump A Priority line shall never be Bumped, and a Priority line shall never Bump another Priority line; among Priority lines the earlier submission wins (BR-15).

Exclusive.Bump.Effect When a line is Bumped the system shall, in one transaction: post a Refund equal to the full Charge (no Refund Schedule applies; a Hold has no Charge and gets none), place the line at the head of its waitlist, notify its Manager with the reason, and if the line was Required in an Allocated Request move that Request to Waiting while its other lines stay allocated (BR-17).

Exclusive.Bump.Audit Every Bump shall be recorded with the Priority line, the Bumped line, the actor who submitted the Priority line, and the timestamp.

4.7 Metered Allocation

4.7.1 Description and Priority

A Metered line asks for a number of units held concurrently over a window, such as four GPUs for two weeks; it is not a bag of hours. A line is satisfiable when, in every hour of its window, the units already allocated or held plus the new units do not exceed the Resource Type's Pool Capacity, and the Team's concurrent units stay within its Quota if one is set. Credits govern affordability; the Quota prevents one credit-rich Team from taking a whole pool. Waitlist rules are the same as for Exclusive lines. Priority lines are Exclusive-only in Release 1; extending them to Metered pools is the Stretch item "service tiers and preemption." This feature realizes FE-7 and is exercised by UC-4 and UC-10. **Priority:** High; the Stretch requirements are Low.

4.7.2 Stimulus/Response Sequences

- **Stimulus:** A Request with a Metered line is evaluated. **Response:** The system checks Pool Capacity and Quota for every hour of the window; the line is held or allocated, or a Required line joins the waitlist.
- **Stimulus:** A Resource Owner reduces Pool Capacity. **Response:** The system refuses a reduction below what is allocated or held in any future hour (Inventory.Pool.Reduce).

4.7.3 Functional Requirements

Metered.Line.Form A Metered line shall name a Resource Type, a number of units to hold concurrently, and a window; the system shall not offer a specific Resource or a Priority mark on a Metered line.

Metered.Pool.Check A Metered line shall be satisfiable only if, for every hour of its window, the units allocated plus the units on Hold for that Resource Type plus the line's units do not exceed the Pool Capacity (BR-20).

Metered.Quota.Check If the Capacity Planner has set a Quota for the requesting Team on the Resource Type, a Metered line shall be satisfiable only if, for every hour of its window, the Team's units allocated plus held plus the line's units do not exceed the Quota (BR-21). With no Quota set there shall be no per-Team cap.

Metered.Quota.Show While drafting, the system shall show the Team's Quota for the Resource Type, if any, and the Team's peak concurrent units already allocated or held inside the line's window.

Metered.Quote The Quote of a Metered line shall be the sum over the hours of its window of units times Unit Price times the Price Band multiplier for that hour (BR-22).

Metered.Waitlist A Required Metered line that is unsatisfiable shall join the waitlist for its Resource Type and window under the rules of Exclusive.Waitlist.Join to Exclusive.Waitlist.View, and shall be satisfied from the waitlist when its units fit in every hour of its window.

Metered.Atomic A Metered line shall be satisfied with its full number of units for its full window or not at all (BR-3).

Metered.Tier.Define [Stretch] The system shall let the Capacity Planner define service tiers for a Metered Resource Type, each with a premium multiple on the Unit Price, so that a Manager can mark a Metered line with a tier in the same way as a Priority line.

Metered.Tier.Bump [Stretch] A line of a higher tier shall be able to Bump units of lower-tier lines under the rules of Exclusive.Priority.Bump to Exclusive.Bump.Effect, the bump cutoff included; the Bumped line shall receive a full Refund and the head of the waitlist.

Metered.Usage.Record [Stretch] The system shall let a Resource Owner record, at the end of a Period, the units actually consumed by each Metered allocation of that Period.

Metered.Usage.Compare [Stretch] The dashboard shall show requested versus used units per Team and Resource Type per Period. Recorded usage shall have no effect on the Ledger.

4.8 Request Lifecycle and Holds

4.8.1 Description and Priority

The three Allocation Classes resolve at different speeds: Exclusive and Metered lines resolve instantly, Matched lines wait for a human approval. A Request must therefore neither lose the room while the people are approved, nor charge for GPUs nobody can use because the people were refused. The lifecycle answers this with Required and Optional lines and with Holds bounded by the Objection Window: when a Request enters Pending, every satisfiable Exclusive and Metered line is placed on Hold, tentative and uncharged, while the Matched lines await approval. The Request is Allocated when every Required line is satisfied, and in that one step all Holds commit and all Charges and Earnings post. If a Required line is refused or the Objection Window lapses, every Hold releases and the Request goes to Waiting. The states and transitions are drawn in Figure 3. This feature realizes FE-8 and underlies UC-4, UC-6, UC-7, and UC-10. **Priority:** High.

4.8.2 Stimulus/Response Sequences

- **Stimulus:** A Request is submitted. **Response:** The system evaluates every line and moves the Request to Allocated, Pending, or Waiting.
- **Stimulus:** The last outstanding Required approval arrives. **Response:** Holds commit, Charges and Earnings post, everyone involved is notified, all in one transaction.

- **Stimulus:** A Required approval is refused, or the Objection Window lapses. **Response:** Every Hold and its held credits release; the Request goes to Waiting; on lapse the outstanding approvals escalate.
- **Stimulus:** Capacity a Waiting Request needs becomes free, or an escalated approval arrives. **Response:** The system checks whether every Required line can now be satisfied at once and, if so, allocates the Request.

4.8.3 Functional Requirements

Lifecycle.States A Request shall be in exactly one of the states Draft, Submitted, Pending, Allocated, Waiting, Withdrawn, or Refused, with exactly the transitions of Figure 3. Allocated, Withdrawn, and Refused are final except that Allocated returns to Waiting when a Required line is Bumped.

Lifecycle.Line.States Each Request Line shall be in exactly one of the states Open, Awaiting approval, Approved, Held, Waitlisted, Allocated, Dropped, Released, Bumped, or Expired; Appendix B defines each.

Lifecycle.Evaluate On submission the system shall, in one transaction, determine for every Exclusive and Metered line whether it is satisfiable now (Exclusive.Conflict, Metered.Pool.Check, Metered.Quota.Check) and for every Matched line whether an approval is needed, and shall move the Request as follows: to Allocated if no line needs approval and every Required line is satisfiable; to Pending if any Required or Optional line needs approval; otherwise to Waiting.

Lifecycle.Hold.Place When a Request enters Pending, the system shall place a Hold on every satisfiable Exclusive and Metered line. A Hold shall block other Requests exactly as an allocation does and shall post no Charge (BR-6).

Lifecycle.Hold.Credits While a line is on Hold, its locked Quote shall count as held credits of the Team, so that concurrent Pending Requests cannot together exceed the Balance (BR-28).

Lifecycle.Hold.Lifetime A Hold shall end in exactly one of these ways: it commits when the Request becomes Allocated; it releases when the Request becomes Waiting, Refused, or Withdrawn; or it releases when the Objection Window lapses. No Hold shall outlive the Objection Window of its Request (BR-6).

Lifecycle.Hold.NoneAfterLapse After a Request's Objection Window has lapsed the system shall place no further Hold for that Request; it shall be allocated only under Lifecycle.Waiting.Reevaluate.

Lifecycle.Window.Start The Objection Window shall start when the Request first enters Pending, shall have the length set by the Capacity Planner (example default two working days), and shall not restart when a Candidate is replaced or the Request re-enters Pending (BR-8).

Lifecycle.Window.Lapse When the Objection Window lapses with a Required approval outstanding, the system shall release every Hold, move the Request to Waiting, and escalate the outstanding approvals (Matched.Escalate), within five minutes of the lapse (PE-4).

Lifecycle.Allocate When every Required line is satisfied, the system shall in one transaction commit every Hold into an allocation, post a Charge equal to the locked Quote for every allocated

Exclusive and Metered line, form the Loans and post the Earnings for approved Matched lines, drop Optional lines under Lifecycle.Optional.Drop, move the Request to Allocated, and notify the Manager (BR-7).

Lifecycle.Optional.Drop At allocation, an Optional line that is unsatisfiable, unapproved, or no longer affordable after the Required lines are charged shall be dropped, never waitlisted, and the Manager shall be told which lines were dropped and why (BR-2).

Lifecycle.Refuse.Line When a Required Matched approval is refused, the system shall release every Hold and its held credits, move the Request to Waiting, and ask the requesting Manager for a replacement (Matched.Refuse.Replace).

Lifecycle.Waiting.Repending When the Manager of a Waiting Request selects a replacement Candidate while the Objection Window is still running, the system shall re-evaluate the Exclusive and Metered lines, place Holds on those now satisfiable, waitlist the Required ones that are not, request the new approval, and move the Request to Pending.

Lifecycle.Waiting.Reevaluate Whenever capacity that a Waiting Request's line waits for frees, or an approval on it arrives, the system shall check whether every Required line of that Request can be satisfied at that moment; if so it shall allocate the Request under Lifecycle.Allocate at the locked Quotes; if not, the freed capacity shall be offered to the next waiting line and the Request's lines shall keep their waitlist positions.

Lifecycle.Refused A Request shall become Refused when a Required Matched line cannot be satisfied under Matched.Refuse.None; every Hold releases and nothing is charged.

Lifecycle.Bump.Return When a Required line of an Allocated Request is Bumped, the Request shall return to Waiting with that line at the head of its waitlist and every other line unchanged (BR-17).

Lifecycle.Withdraw The Manager shall be able to withdraw a Request from Draft, Pending, or Waiting at any time (Draft.Withdraw).

Lifecycle.Audit Every change of Request state or line state shall be recorded with the actor (a Member, or the system for timed events), the change, and the timestamp, and the record shall be visible to the Request's Manager and to the Capacity Planner (BR-9).

Lifecycle.Notify The system shall send an in-app notification to the affected Manager when a Request enters Pending, Allocated, Waiting, or Refused, when a Hold expires, when a line is Bumped or satisfied from the waitlist, and when a Request escalates; to a lending Manager when an approval is requested; and to the Capacity Planner when a Request escalates (CI-2).

4.9 Release and Refund

4.9.1 Description and Priority

A Manager who no longer needs an allocated Exclusive or Metered line, or part of it, gives it back with a Release. Credits are refunded according to the Resource Type's Refund Schedule: a step function of how far ahead of the window start the Release happens, capped below 100% so that reserving is never free. A partial refund taxes the over-reservation itself while still rewarding early Release over a silent no-show, and is the first concrete mechanism behind the proposal's claim that

the system accounts for the failure modes of a simple market. Ending a Loan early prorates the lending Team's Earning; no Refund Schedule applies because nothing was charged. This feature realizes FE-9 and is exercised by UC-9. **Priority:** High.

4.9.2 Stimulus/Response Sequences

- **Stimulus:** The Manager Releases an allocated line in whole or in part. **Response:** The system posts the Refund due under the Refund Schedule, frees the capacity, and offers it to the waitlist.
- **Stimulus:** The borrowing Manager ends a Loan early. **Response:** The system shortens the Loan and posts an Earning correction to the lending Team for the days not lent.

4.9.3 Functional Requirements

Release.Line The system shall let the Manager of the owning Team Release an allocated Exclusive or Metered line in whole or in part before it is consumed. A partial Release shortens the window, removes occurrences, or reduces units; the remaining part keeps its allocation, its instance, and its locked unit prices.

Release.Refund On Release the system shall post one Refund equal to the Charge attributable to the released part multiplied by the fraction from the Resource Type's Refund Schedule for the lead time between the Release and the start of the released part (BR-25).

Release.Schedule.Form A Refund Schedule shall be a list of steps, each a minimum lead time and a fraction; the fraction for a Release is that of the largest step whose minimum lead time is not exceeded by the actual lead time. Example default: 75% at seven or more days ahead, 25% at less than seven days but before the window starts, 0% once the window has started.

Release.Schedule.Cap The system shall refuse a Refund Schedule containing a fraction of 100% or more (BR-25).

Release.Bucket A Refund shall return credits to the same part of the Account, granted or earned, from which the Charge was drawn, in the same proportions (BR-26).

Release.Waitlist Capacity freed by a Release shall immediately be offered to the waitlist under Exclusive.Waitlist.Fill.

Release.Loan.End The system shall let the borrowing Manager end a Loan early from a chosen date. The lending Manager shall be notified.

Release.Loan.Prorate When a Loan ends early, the system shall post to the lending Team an Earning correction equal to minus the Day Rate times the person-days no longer lent, so that the Team's Earning for the Loan equals the Day Rate times the days actually lent. No Refund Schedule applies (BR-11). This correction is the only Transaction that may take a Balance below zero; while the Balance is below zero the Team can submit no Request with a Required Exclusive or Metered line (BR-29).

Release.Idle Capacity that is allocated but neither used nor Released shall appear on the dashboard as idle (Dashboard.Idle); the system shall not release it automatically.

Release.Audit Every Release shall be recorded with the line, the released part, the lead time, the fraction applied, the Refund, the actor, and the timestamp.

4.10 Ledger and Pricing

4.10.1 Description and Priority

The Ledger records every Team's Account and every Transaction; it holds no capacity information, which lives in the Inventory. Exclusive and Metered lines cost credits; Matched lines cost nothing, and lending a Person earns the lending Team a flat Day Rate. Each Team's Account has granted credits, which the Capacity Planner grants each Period and which expire at its end, and earned credits, which carry over indefinitely; this two-part Account is what keeps Managers lending late in a Period. Prices vary by Price Band, a time-of-day and day-of-week multiplier on the Unit Price, which rewards use at low-demand times and is the second concrete mechanism behind the proposal's market claim. Quotes are locked at submission. This feature realizes FE-10 and is exercised by UC-11 and UC-12. **Priority:** High; the Stretch requirement is Low.

4.10.2 Stimulus/Response Sequences

- **Stimulus:** A Period starts. **Response:** The system posts each Team's Grant, expires last Period's unspent granted credits, and opens a new Statement.
- **Stimulus:** A Request becomes Allocated. **Response:** Charges post to the requesting Team at the locked Quotes and Earnings post to each lending Team.
- **Stimulus:** A Manager opens the Statement. **Response:** The system shows the Period's Grants, Charges, Earnings, Refunds, what expired, and every Transaction behind them.

4.10.3 Functional Requirements

Ledger.Account Each Team shall have exactly one Account holding three amounts in credits: granted, earned, and held. Balance shall equal granted plus earned; Available Balance shall equal Balance minus held.

Ledger.Transaction.Kinds Every Transaction shall be of exactly one of four kinds: Grant, Charge, Earning, or Refund, and shall record the Team, the amount in credits, the Period, the posting timestamp, the actor or system job that caused it, and the Request Line it concerns where there is one (BR-30).

Ledger.AppendOnly Transactions shall never be updated or deleted. A correction shall be a new Transaction that references the one it corrects (QA-3).

Ledger.Held Held credits shall equal the sum of the locked Quotes of the Team's lines currently on Hold, and shall change only when a Hold is placed, commits, or releases.

Ledger.Grant.Period At the start of each Period the system shall post to every Team a Grant of the amount the Capacity Planner has set for that Team and Period (Planner.Settings).

Ledger.Grant.Adjust The Capacity Planner shall be able to post a mid-Period Grant to a Team with a reason, for example to settle an escalation; the amount is added to granted credits and expires with them.

Ledger.Expire At the end of a Period the system shall set every Team's granted credits to zero, record the expired amount on the Statement, post the new Grant, and leave earned credits untouched (BR-27). The rollover shall be effective within fifteen minutes of the Period boundary (PE-4).

Ledger.SpendOrder A Charge shall draw from granted credits first and from earned credits only when granted credits are exhausted, and shall record how much it drew from each (BR-26).

Ledger.Charge.Post A Charge shall post only when the Request becomes Allocated and shall equal the line's locked Quote (BR-7, BR-24).

Ledger.Earning.Post An Earning shall post to the lending Team when the Loan forms, equal to the Day Rate times the person-days of the Loan, into earned credits (BR-11). A person-day is one Monday-to-Friday calendar day inside the Loan's window.

Ledger.NoNegative The system shall never post a Charge that would take a Team's Balance below zero, shall offer no borrowing, and shall move no credits between Teams (BR-29).

Ledger.Price.Unit The Capacity Planner shall set for each Exclusive and Metered Resource Type a Unit Price in credits: per instance-hour for Exclusive types and per unit-hour for Metered types. Matched types have no Unit Price (BR-10).

Ledger.Price.Band The Capacity Planner shall define for each Exclusive and Metered Resource Type zero or more Price Bands, each a set of weekdays, a start and end time of day, and a multiplier; bands of one type shall not overlap; hours in no band have multiplier 1.0 (BR-23).

Ledger.Price.Formula The Quote of an Exclusive or Metered line shall be the sum over each hour of each occurrence of quantity times Unit Price times the multiplier of the Price Band containing that hour, multiplied by the priority premium if the line is a Priority line (BR-22).

Ledger.Price.Premium The priority premium shall be one multiple set by the Capacity Planner and applied to every Priority line (example default 2×; BR-15).

Ledger.Quote.Lock A Quote shall be locked at submission and shall not change afterward; a change to any price setting shall affect only lines not yet submitted (BR-24).

Ledger.DayRate The Capacity Planner shall set one Day Rate in credits per person-day, identical for every Person regardless of skill (BR-11).

Ledger.Statement For each Team and Period the system shall produce a Statement showing opening granted and earned credits, every Grant, Charge, Earning, and Refund with its Transaction, the amount expired at Period end, and closing granted and earned credits; a Manager shall see their own Team's Statements and the Capacity Planner every Team's.

Ledger.Period The Period shall be a fixed calendar interval set by the Capacity Planner (example default one calendar month); the system shall refuse a change that would shorten the current Period to end in the past (BR-31).

Ledger.Surcharge [Stretch] The Capacity Planner shall be able to enable a demand surcharge per Resource Type: an additional multiplier on the Unit Price that rises with the fraction of the type's capacity already allocated or held in the requested window at quote time, applied in Ledger.Price.Formula and locked with the Quote.

4.11 Showback Dashboard

4.11.1 Description and Priority

The dashboard makes capacity and credits visible without any billing: utilization and idle capacity per Resource Type, and credits earned and spent per Team. Managers see their own Team against the company-wide utilization; the Capacity Planner sees every Team; Resource Owners see their own Resources. It is the showback artifact promised in the proposal, together with the Statement. This feature realizes FE-11 and is exercised by UC-12. **Priority:** Medium.

4.11.2 Stimulus/Response Sequences

- **Stimulus:** A Manager opens the dashboard. **Response:** The system shows the current Period's utilization and idle capacity per Resource Type and the Team's credits earned, spent, and remaining.
- **Stimulus:** The Capacity Planner selects a past Period. **Response:** The system shows the same figures for that Period for every Team.

4.11.3 Functional Requirements

Dashboard.Utilization For each Exclusive and Metered Resource Type and Period the dashboard shall show utilization: allocated instance-hours divided by available instance-hours for Exclusive types, and allocated unit-hours divided by Pool Capacity times hours for Metered types, both as a percentage.

Dashboard.Idle For each Exclusive and Metered Resource Type the dashboard shall show idle capacity: available minus allocated, for the current Period to date and for the coming seven days, so that free capacity is visible before it is wasted.

Dashboard.Credits For each Team and Period the dashboard shall show Grants received, Charges, Earnings, Refunds, credits expired, and the current Balance and Available Balance, with a link to the Statement.

Dashboard.Loans For each Team and Period the dashboard shall show person-days lent to other Teams and person-days borrowed.

Dashboard.Waitlist For each Exclusive and Metered Resource Type the dashboard shall show the number of lines waiting and the oldest waiting submission time.

Dashboard.Scope A Manager shall see company-wide utilization and idle capacity and only their own Team's credits and Loans; the Capacity Planner shall see every Team; a Resource Owner shall see utilization and idle capacity for the Resources they own.

Dashboard.Period The dashboard shall default to the current Period and shall let the viewer select any past Period.

Dashboard.Fresh Dashboard figures shall reflect every allocation, Release, and Transaction completed before the page was requested.

Dashboard.Usage [Stretch] The dashboard shall show requested versus used units per Team and Metered Resource Type when usage has been recorded (Metered.Usage.Compare).

4.12 Administration

4.12.1 Description and Priority

An Administrator creates Members and Teams, grants roles, maintains the skill list, and defines Resource Types (Section 4.1). The Administrator configures the system and sets no policy. Policy settings, the Grants, prices, and windows that shape behavior, belong to the Capacity Planner and are collected here so that every configurable value in this document has one home. This feature realizes FE-12 and is exercised by UC-1 and UC-11. **Priority:** High.

4.12.2 Stimulus/Response Sequences

- **Stimulus:** An Administrator creates a Member and assigns a Team and roles. **Response:** The Member can log in and sees only the functions of their roles.
- **Stimulus:** The Capacity Planner changes a setting. **Response:** The system applies it to future Quotes, Grants, or windows, records the change, and never alters a locked Quote.

4.12.3 Functional Requirements

Admin.Member.Create An Administrator shall be able to create a Member with a name, email address, username, initial password, and Team, and to deactivate a Member. A deactivated Member cannot log in, appears on no Shortlist, and keeps their history.

Admin.Role.Assign An Administrator shall be able to grant and revoke the roles Manager, Resource Owner, Capacity Planner, and Administrator for any Member. Each Team shall have exactly one Manager at any time; assigning a new Manager replaces the previous one.

Admin.Team.Create An Administrator shall be able to create, rename, and retire a Team, and to move a Member between Teams. A retired Team can submit no Request; its Account and history remain readable.

Admin.Skill.List An Administrator shall maintain the list of skills from which Members choose; a skill in use on any profile or line may be renamed but not deleted.

Admin.Login The system shall authenticate every user by username and password and shall let a Member change their own password. Passwords shall be stored hashed (SE-2).

Admin.NoPolicy Administrator screens shall contain no price, Grant, Quota, or window setting; those settings shall be available only to the Capacity Planner (BR-32).

Admin.Settings.Log Every change to a setting by an Administrator or the Capacity Planner shall be recorded with the actor, the old and new value, and the timestamp.

Planner.Settings The Capacity Planner shall be able to set, as data and without deployment: the Period (default one calendar month); each Team's Grant for each coming Period; the Unit Price of each Exclusive and Metered Resource Type; the Price Bands of each such type (example defaults: weekdays 09:00–17:00 ×1.5, all other hours ×0.5); the Day Rate; the Refund Schedule of each such type (example default 75%, 25%, 0%); the Quota of any Team on any Metered type; the priority premium (default 2×); the bump cutoff of each Exclusive type (default 24 hours); and the length of the Objection Window (default two working days).

Planner.Settings.Effect A changed price setting shall affect Quotes not yet locked; a changed Grant shall affect Periods not yet started unless posted as a mid-Period Grant; a changed Objection Window, bump cutoff, or Refund Schedule shall affect events after the change. No change shall alter a posted Transaction.

Planner.Escalations The Capacity Planner shall see a list of escalated Requests with the outstanding approvals, the silent Manager, and the time since the Objection Window lapsed, and shall decide each approval from that list (Matched.Escalate).

5 External Interface Requirements

5.1 User Interfaces

The user interface is a browser application with one navigation menu whose entries depend on the Member's roles. Every screen shows the Member's name, Team, and a notification indicator. Field-level help is available for every setting and every column of the Draft Request screen (UD-3). The key screens are:

UI-1 The application shell shall show a role-dependent menu (Requests, Suggestions, Approvals, Inventory, Dashboard, Statement, Settings, Administration), the signed-in Member and Team, and an in-app notification list with unread count.

UI-2 The Draft Request screen shall show the Request's lines as a table with, per line, Resource Type, Allocation Class, quantity, window, recurrence, Required or Optional, Priority mark where offered, and Quote; totals for Required and Optional lines; the Team's Available Balance and any shortfall; a panel of Open Suggestions that can be accepted into lines; and the actions Save, Split line, Submit, and Withdraw.

UI-3 The Request detail screen shall show the Request's state, the Objection Window end when Pending, each line with its state, instance, Hold or waitlist position, locked Quote, and the complete state history with actor and timestamp.

UI-4 The Shortlist screen shall list Candidates in Shortlist order with name, Team, skills, and person-days lent in the current Period, with a selection control limited to the line's quantity.

UI-5 The Approvals screen shall list approval requests addressed to the signed-in Manager with requesting Team, Person, window, Earning, and time left in the Objection Window, with Approve and Refuse actions and a reason field.

UI-6 The Inventory screens shall list Resource Types and Resources with filters, and shall show for a chosen Resource a timeline of Holds and allocations; Resource Owners shall have the add, edit, retire, and units controls for their own Resources.

UI-7 The Profile screen shall show and let the Member edit their skills and availability ranges and shall list their Loans; a Manager's Team roster shall show the same for each Member of the Team.

UI-8 The Dashboard screen shall show the figures of Section 4.11 with a Period selector, as tables and bar charts.

UI-9 The Statement screen shall show a Team's Statement for a chosen Period with the Transaction list beneath it.

UI-10 The Settings screen, visible to the Capacity Planner only, shall show every setting of Planner.Settings with its current value, example default, and change history, and the list of escalated Requests.

UI-11 The Administration screens, visible to Administrators only, shall manage Members, Teams, roles, the skill list, and Resource Types.

5.2 Hardware Interfaces

The system has no hardware interfaces. It controls no door, instrument, or compute node; a Loan, an Exclusive allocation, or a Metered allocation is a record that people act on.

5.3 Software Interfaces

SI-1 The system shall store all persistent data in a PostgreSQL 16 database through JDBC; the schema shall be created and migrated by the application at start-up. Automated tests shall run against PostgreSQL 16 in a container started by the test suite (Testcontainers) with the same migrations.

SI-2 The front end shall communicate with the back end only through a REST/JSON API over HTTPS. Every use case in Appendix D shall be achievable through that API with the same role enforcement as the user interface, so that automated tests and a future integration can drive the system.

SI-3 The back end shall run on Java 17 with Spring Boot, packaged as one Docker image that contains the built front end.

SI-4 The front end shall run in the current release of Chrome, Firefox, Safari, and Edge on desktop without plug-ins.

SI-5 Release 1 shall have no interface to any calendar, HR, identity, or cluster-scheduling system (LI-3).

5.4 Communications Interfaces

CI-1 All traffic between browser and server shall use HTTPS; the server shall accept no plain HTTP connection except a redirect to HTTPS.

CI-2 All notifications in Release 1 shall be delivered inside the application, listed newest first and marked read by the Member; no notification shall leave the system.

CI-3 [Stretch] The system shall optionally send each in-app notification also by email through an SMTP server configured by the Administrator.

6 Other Nonfunctional Requirements

6.1 Performance Requirements

PE-1 The system shall support 500 Members, 50 Teams, 200 Resources, and 10,000 Requests per year without violating PE-2 or PE-3.

PE-2 Any page shall be served in at most 2 s at the 95th percentile with 50 concurrent users, at the scale of PE-1.

PE-3 Building a Shortlist against 500 Members shall take at most 3 s.

PE-4 Objection Window lapse, Hold expiry, and waitlist expiry shall take effect within 5 min of their due time.

PE-5 Period rollover (expiring granted credits and posting the new Grants) shall take effect within 15 min of the Period boundary.

6.2 Safety Requirements

None. The system controls no physical equipment and its records cannot cause injury or damage; an incorrect allocation is corrected by a Release or a Bump.

6.3 Security Requirements

SE-1 The server shall enforce the signed-in Member's roles on every action, whether it arrives from the user interface or directly through the API; the user interface hiding a control shall never be the only protection.

SE-2 Passwords shall be stored only as salted hashes produced by an adaptive algorithm such as bcrypt; no password shall be logged or shown.

SE-3 All communication shall be encrypted in transit (CI-1).

SE-4 The system shall store no personal data beyond name, email address, Team, skills, and availability.

SE-5 A Member shall see only what their roles allow: their own profile and Loans; as Manager, their Team's roster, Requests, and Statements and the Shortlists of their own Requests; as Capacity Planner, every Team's; as Resource Owner, the timelines of their Resources. Statements of other Teams shall never be visible to a Manager.

SE-6 Sessions shall expire after 12 hours of inactivity and every authentication attempt, success or failure, shall be logged with timestamp and username.

6.4 Software Quality Attributes

QA-1 **Correctness.** Under concurrent submissions the system shall never double-book an Exclusive instance for overlapping windows and shall never allocate a Metered Resource Type beyond its Pool Capacity in any hour. Conflict detection and allocation shall be serialized per Resource Type in the database.

QA-2 **Atomicity.** Holds shall commit and the corresponding Charges and Earnings shall post in one transaction or not at all; a failure part-way shall leave no Hold committed and no Transaction posted (BR-7).

QA-3 **Auditability.** Ledger Transactions shall be append-only, and every Request and line state change shall record actor, change, and timestamp (BR-9).

QA-4 **Availability.** The system shall be available during business hours; planned maintenance shall be scheduled outside them. There is no high-availability requirement: a single server with a nightly database backup is sufficient for Release 1.

QA-5 **Usability.** An untrained Manager shall be able to draft and submit a three-line Request in under 5 min. This shall be tested with at least 3 people from outside the team before the final demonstration, and the results recorded in the inspection report.

QA-6 **Portability.** The user interface shall work in the current release of Chrome, Firefox, Safari, and Edge; no mobile layout is required.

QA-7 **Extensibility.** Adding a Resource Type shall require no code. Adding an Allocation Class shall require implementing one allocator interface and no change to the existing allocators, the Request lifecycle, or the Ledger.

QA-8 **Data retention.** Requests, their state histories, Suggestions, and Transactions shall be kept for at least 2 years; nothing is purged in Release 1.

QA-9 **Testability.** Every business rule in Section 6.5 shall be covered by at least one automated test that runs against PostgreSQL in a container started by the test suite, so that the allocation and Ledger rules can be checked on every build.

6.5 Business Rules

The rules below govern behavior across features. Each is referenced by number from the requirements and use cases that depend on it.

Requests and lines

BR-1 Only a Manager creates, submits, edits, or withdraws a Request, and only on behalf of their own Team. Members express needs as Suggestions to their own Manager.

BR-2 Every Request Line is Required (default) or Optional. A Request is Allocated only when every Required line is satisfied. Optional lines allocate if they can, never block the outcome, and are dropped rather than waitlisted when they cannot.

BR-3 A Request Line is atomic in every Allocation Class: it is satisfied with its full quantity, for its full window, and for every occurrence of a recurrence, or not at all. A Manager who will accept less splits the line.

BR-4 Every Resource has exactly one Resource Type; every Resource Type has exactly one Allocation Class; every Request Line takes its Allocation Class from its Resource Type.

BR-5 A Matched line either lists skills with a quantity, or names one Person with quantity one. A named Person is checked only for availability and for a Loan overlapping the window; both forms need the lending Manager's approval.

Holds and the Objection Window

BR-6 A Hold is a tentative, uncharged allocation placed while a Request's Matched lines await approval. It blocks other Requests as an allocation does. It lives at most until the Request's Objection Window lapses, and it releases whenever the Request leaves Pending without being Allocated. No Hold is placed for a Request whose Objection Window has lapsed.

BR-7 When a Request becomes Allocated, all of its Holds commit and all of its Charges and Earnings post in one transaction. Nothing posts while a Request is Pending or Waiting.

BR-8 The Objection Window is the period, set by the Capacity Planner (example default two working days), that lending Managers have to approve or refuse. It starts when the Request first enters Pending, belongs to the Request, and does not restart when a Candidate is replaced. When it lapses, outstanding approvals escalate to the Capacity Planner, who decides on the silent Manager's behalf, and every later approval on that Request goes to the Capacity Planner directly.

BR-9 Every change of a Request's or line's state records the actor, the change, and the timestamp. Timed changes record the system as actor.

Matched lines

BR-10 Matched lines have no Unit Price and cost the requesting Team nothing.

BR-11 A Loan earns the lending Team the Day Rate for every person-day of the Loan, the same for every Person regardless of skill, posted when the Loan forms. If the Loan ends early the Earning is corrected to the days actually lent; no Refund Schedule applies.

BR-12 A Manager cannot approve lending to their own Team, and Members of the requesting Team are never Candidates for it. Work inside a Team is not a Loan.

BR-13 A Candidate for a skill-based line holds every listed skill, is available for the whole window, and has no Loan overlapping any day of it: a Person is in at most one Loan at a time. Availability ranges are the Member's own and a Loan never changes them; a Loan that ends early frees the Person from the day after its new last day. The Shortlist is ordered by fewest person-days lent in the current Period, counting every formed Loan, ties by name. The requesting Manager selects; the system never selects automatically. Each selected Candidate needs one approval from their own Manager.

BR-14 A Person is notified when selected and when their Loan forms, and cannot decline in Release 1.

Exclusive lines, Priority, and the waitlist

BR-15 A Priority line is an Exclusive line whose Quote is multiplied by the priority premium

set by the Capacity Planner (example default 2×). A Priority line can Bump standard lines, cannot be Bumped, and never Bumps another Priority line; among Priority lines the earlier submission wins.

BR-16 No Bump happens inside the bump cutoff, set per Exclusive Resource Type by the Capacity Planner (example default 24 hours), before the Bumped line's window or occurrence starts.

BR-17 A Bumped line goes to the head of the waitlist for its Resource Type and window, its Manager is notified, and its Charge is refunded in full because the Team did nothing wrong. If it was a Required line of an Allocated Request, that Request returns to Waiting while its other lines stay allocated.

BR-18 Waitlists exist per Resource Type and window. Required lines that cannot be satisfied join them; Optional lines never do. Order is Priority lines first, then submission time of the Request; a Bumped line goes to the head. A waiting line expires when its window start passes; its Request stays Waiting until edited or withdrawn.

BR-19 Priority lines are offered for Exclusive lines only in Release 1. Extending them to Metered lines is the Stretch item "service tiers and preemption."

Metered lines

BR-20 In every hour of a Metered line's window, the units already allocated plus the units on Hold plus the line's units may not exceed the Resource Type's Pool Capacity.

BR-21 A Quota is an optional per-Team cap on concurrent units of one Metered Resource Type, set by the Capacity Planner. Where set, the Team's allocated plus held plus requested units may not exceed it in any hour. Unset means uncapped.

Credits and prices

BR-22 The Quote of an Exclusive or Metered line is the sum over each hour of each occurrence of quantity × Unit Price × Price Band multiplier, times the priority premium for a Priority line. For a Metered line the quantity is its units; for an Exclusive line its instances.

BR-23 Price Bands are defined per Exclusive and Metered Resource Type by the Capacity Planner as non-overlapping weekday and time-of-day windows with a multiplier; hours in no band have multiplier 1.0 (example defaults: weekdays 09:00–17:00 ×1.5, all other hours ×0.5). The Day Rate is never banded.

BR-24 The Quote is shown while drafting and locked at submission. The Charge that posts equals the locked Quote, and held credits equal the locked Quote. Price changes affect only lines not yet submitted.

BR-25 On Release, the Refund is the Charge for the released part times a fraction from the Resource Type's Refund Schedule, a step function of the lead time to the released part's start (example default: 75% at seven or more days ahead, 25% at less than seven days ahead, 0% after the start). No step is 100% or more.

BR-26 A Charge draws granted credits first, then earned credits. A Refund returns credits to the part they were drawn from, in the same proportions.

BR-27 Granted credits expire at the end of the Period in which they were granted. Earned credits carry over indefinitely. At rollover the system sets granted credits to zero, records the expired amount on the Statement, and posts the new Grant.

BR-28 Available Balance equals granted plus earned minus held credits. A Request can be submitted only if the sum of the locked Quotes of its Required lines does not exceed the Team's Available Balance at submission. Optional lines that no longer fit at allocation are dropped.

BR-29 No Charge may take a Team's Balance below zero. There is no borrowing and no transfer of credits between Teams. The only Transaction that may take a Balance below zero is an Earning correction after a Loan ends early; until a Grant or Earning restores it, that Team can submit no Request with a Required Exclusive or Metered line.

BR-30 The Ledger holds exactly four kinds of Transaction: Grant, Charge, Earning, and Refund. Transactions are append-only; a correction is a new Transaction that references the one it corrects.

BR-31 The Period is a fixed calendar interval set by the Capacity Planner (example default one calendar month). Grants are issued at its start and expire at its end.

Roles

BR-32 Roles are enforced on the server for every action. The Capacity Planner and the Administrator are distinct roles that one Member may hold together; the Administrator sets no policy, the Capacity Planner administers no users, and a Resource Owner never approves an allocation.

A Glossary

An internal marketplace where a team's manager posts one request for people, exclusive resources, and metered capacity, and the system allocates each part under the policy that fits its allocation class, settled in credits.

Naming rule. prefer the everyday word a Manager would use. Credit is the unit and never a transaction direction. No money words (bank, funds, pay, debit), because credits are showback, not billing.

Each entry gives the term, its definition, and the words this document deliberately avoids for the same idea. The glossary is generated from the project's `CONTEXT.md` at build time, so the document and the team's working vocabulary cannot drift apart.

A.1 People and roles

Member Any user of the system. Belongs to exactly one Team, has a skill profile and availability, and is lendable as a Person. *Avoid:* user, employee, staff.

Team The organizational unit that owns an Account and every Request made on its behalf. Has exactly one Manager. *Avoid:* group, department, org.

Manager The Member who leads a Team. The only actor who submits Requests, and the approver for lending that Team's Members. *Avoid:* team lead, owner, approver.

Resource Owner A Member responsible for the Inventory records of specific Exclusive or Metered resources. Never approves allocations. *Avoid:* admin, custodian.

Capacity Planner The role that sets each Team's Grant per Period, Unit Prices, the Day Rate, and Refund Schedules, and receives escalations. *Avoid:* planner, finance, admin.

Administrator The role that manages users, Teams, and Resource Type definitions. Configures the system; sets no policy. *Avoid:* superuser, root.

A.2 Resources

Resource Anything that can be allocated: a Person, an Exclusive resource such as a room, or a Metered pool such as GPU hours. Every Resource has exactly one Resource Type. *Avoid:* asset, item, capacity (as a noun for a thing).

Resource Type A named definition, created as data by an Administrator, that fixes a Resource's Allocation Class and attributes. *Avoid:* descriptor, category, kind.

Allocation Class One of three policies a Resource Type declares, deciding how Requests for it are allocated: Matched, Exclusive, or Metered. *Avoid:* resource class, policy type.

Matched The Allocation Class for People. Allocated by matching skills and availability, gated by the Person's Manager. *Avoid:* matched agents, agents, talent.

Person A Member viewed as a Matched Resource: someone with skills and availability who can be lent to another Team. *Avoid:* agent, worker, headcount.

Exclusive The Allocation Class for Resources held whole over a time window, such as rooms or instruments. Allocated with conflict detection, a waitlist, and priority bumping. *Avoid:* reservable, bookable.

Metered The Allocation Class for divisible Resources drawn against a Team quota, such as compute hours or license seats. *Avoid:* pooled, consumable, quota resource.

Pool Capacity The total units a Metered Resource Type can supply at once, such as 32 GPUs, maintained by its Resource Owner. Allocations in any hour may not exceed it. *Avoid:* quota, limit, size.

Quota An optional per-Team cap on the units of one Metered Resource Type the Team may hold concurrently, set by the Capacity Planner. A safety valve against monopolizing a pool, not a rationing mechanism. *Avoid:* budget, allowance, share, capacity.

Inventory The record of every Resource that exists, its Resource Type, and who currently holds it or has it on Hold. *Avoid:* capacity ledger, catalog, registry.

A.3 Requests

Request A single request, owned by a Team and submitted by its Manager, containing one or more Request Lines that may span all three Allocation Classes. *Avoid:* composite request, order, booking, ticket, bundle.

Request Line One need inside a Request: a Resource Type, a quantity, and a time window. Every line has exactly one Allocation Class, taken from its Resource Type. *Avoid:* item, line item, sub-request.

Suggestion A single Request Line a Member proposes to their own Manager. Has no allocation effect until the Manager accepts it into a Request; several Suggestions may be accepted into the same line. *Avoid:* draft, proposal, wish.

Required line / Optional line A Manager's marking on each Request Line. A Request is fulfilled only when every Required line is allocated; Optional lines allocate if they can and never block the outcome. *Avoid:* mandatory, nice-to-have, hard/soft.

Hold A tentative, uncharged allocation of an Exclusive or Metered line placed while a Request's Matched lines await approval. Commits when the Request is fulfilled, releases when it fails or the Objection Window lapses. *Avoid:* reservation, soft booking, lock.

Objection Window The bounded period a lending Manager has to approve or refuse a Matched line before the Request escalates to the Capacity Planner. Also the lifetime of any Hold on that Request. *Avoid:* SLA, approval deadline, timeout.

Candidate A Person who satisfies a Matched line's skills and availability, has no Loan overlapping the line's window, and appears on its Shortlist. Becomes a Loan once their Manager approves. *Avoid:* match, applicant, option.

Shortlist The list of Candidates the system produces for a skill-based Matched line, ordered by fewest person-days lent in the current Period, from which the requesting Manager selects. *Avoid:* search results, pool, matches.

Waitlist The ordered list of Required lines that could not be satisfied for a Resource Type and window. Priority lines first, then by submission time; a Bumped line goes to the head. Optional lines are never waitlisted. *Avoid:* queue, backlog, pending list.

Loan A Person allocated to a Team other than their own for a time window. The fulfilled form of a Matched line. A Person is in at most one Loan at a time. *Avoid:* assignment, secondment, transfer, placement.

Priority line An Exclusive Request Line that is charged a premium, a multiple of the Unit Price set by the Capacity Planner. Can Bump a standard line and cannot itself be Bumped. *Avoid:* urgent, high-priority flag, VIP.

Bump The displacement of a standard Exclusive line by a Priority line. The displaced line goes to the head of the waitlist with a full Refund. Not allowed inside the bump cutoff before the window starts. *Avoid:* preempt, evict, override.

Release A Manager giving back an allocated Exclusive or Metered line, in whole or in part, before

it is consumed, or ending a Loan early. Triggers a Refund according to the Refund Schedule. *Avoid:* cancel, return, un-book.

A.4 Credits

Credit The unit of account in the Ledger. No real money; used for showback, not billing. *Avoid:* point, token, dollar, funds, money.

Ledger The record of every Team's Account and every Transaction. Holds no capacity information. *Avoid:* credit ledger, capacity ledger, bank, wallet.

Account A Team's standing in the Ledger: its granted credits, its earned credits, and its held credits. *Avoid:* wallet, budget (as the thing itself), balance sheet.

Balance A Team's granted credits plus earned credits. *Avoid:* funds, total.

Available Balance Balance minus held credits. What a Team can commit to a new Request. *Avoid:* free balance, spendable.

Held credits Credits set aside for a Request that is Pending, equal to what its Holds will cost when they commit. *Avoid:* encumbrance, reserved credits, pending charges.

Transaction One entry in the Ledger against an Account. Exactly four kinds: a Grant (from the Capacity Planner), a Charge (for an allocated Exclusive or Metered line), an Earning (for a Loan), or a Refund (for a Release). *Avoid:* debit, credit (as a direction), payment, entry.

Grant Credits given to a Team by the Capacity Planner at the start of a Period, or as a mid-period adjustment. Expire at the end of the Period. *Avoid:* allowance, budget allocation, deposit.

Period The fixed calendar interval over which Grants are issued and expire. Set by the Capacity Planner. *Avoid:* cycle, quarter, budget period.

Statement A Team's per-Period view of its Account: Grants, Charges, Earnings, Refunds, and what expired. The showback artifact. *Avoid:* report, bill, invoice.

Unit Price The base credits per unit of an Exclusive or Metered Resource Type (per hour of a room, per GPU hour), set by the Capacity Planner, before any Price Band multiplier. Matched lines have no Unit Price. *Avoid:* cost, rate, fee, base price.

Price Band A time-of-day and day-of-week window, defined per Resource Type by the Capacity Planner, carrying a multiplier on the Unit Price. Rewards use at low-demand times. *Avoid:* tariff, peak/off-peak (as terms), surge.

Quote The credits a Request Line will cost, shown to the Manager while drafting and locked when the Request is submitted. The Charge that posts equals the Quote. *Avoid:* estimate, price tag, invoice.

Day Rate The flat credits a Team earns per person-day of a Loan, set once by the Capacity Planner. The same for every Person regardless of skill. *Avoid:* lending rate, headcount rate, salary, chargeback.

Refund Schedule The rule, set per Resource Type by the Capacity Planner, giving the fraction of a Charge returned on Release as a step function of how far ahead of the window start the Release happens. Never reaches 100%. *Avoid:* cancellation policy, penalty.

B Data Dictionary and Data Model

B.1 Data Model

Figure 2 shows the entities the requirements name and the relationships between them. Attributes are the ones that appear in requirements; the data dictionary below gives their meaning and type. The Inventory is the Resource Type and Resource entities together with the Holds and allocations on them; the Ledger is the Account and Transaction entities. Neither holds the other's information (BR-30).

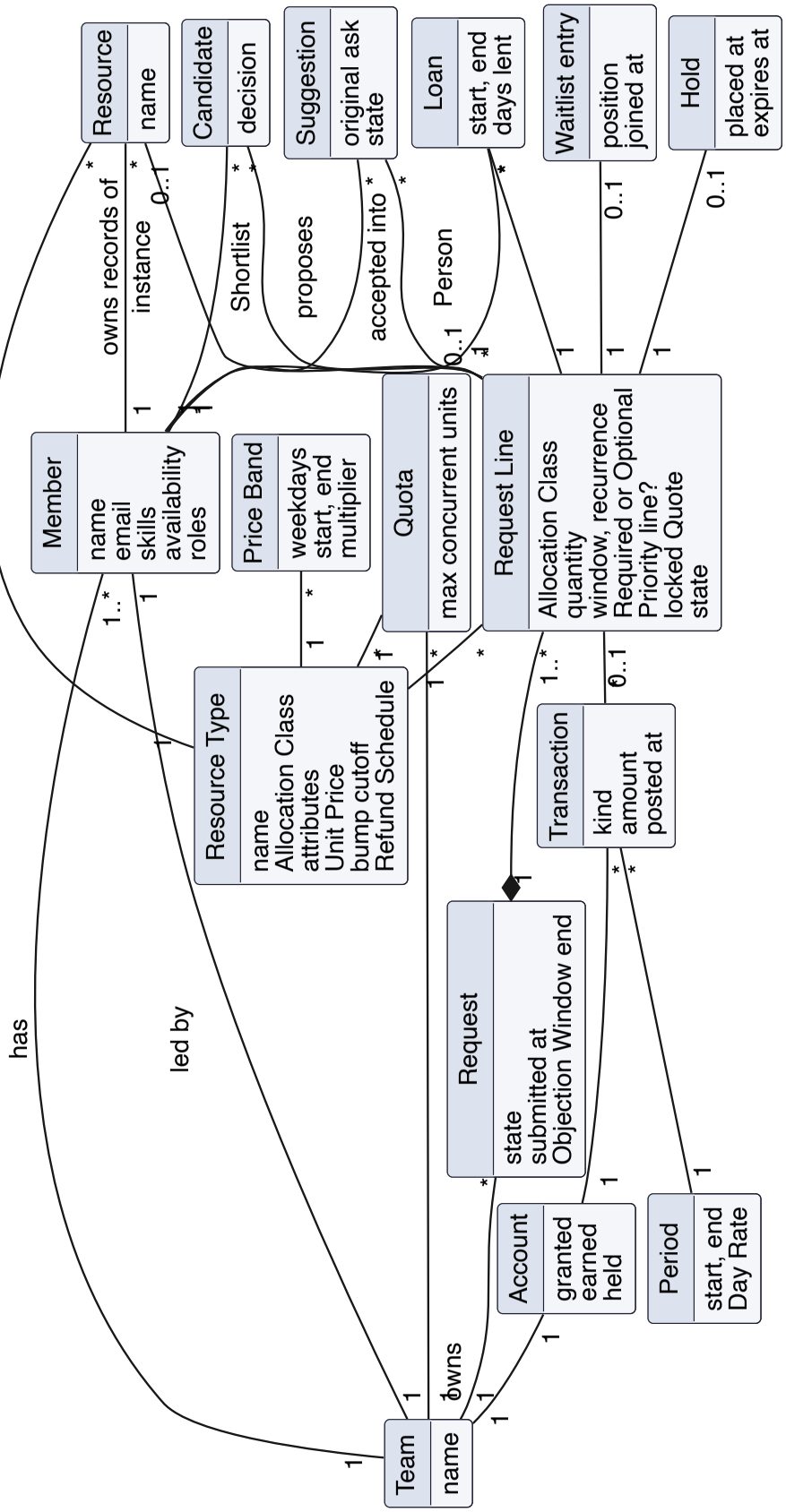


Figure 2: Data model. Multiplicities read along each line; a Statement is a per-Period view over Transactions and is not stored.

B.2 Data Dictionary

Element	Meaning	Composition and type
Team	The organizational unit that owns an Account and its Requests.	name (text, unique); Manager (one Member); retired (yes or no)
Member	Any user; belongs to one Team; lendable as a Person.	name; email (text, unique); username; password hash; Team; skills (set of Skill); availability (set of date ranges); roles (subset of Manager, Resource Owner, Capacity Planner, Administrator); active (yes or no)
Skill	An entry of the Administrator-maintained skill list.	name (text, unique)
Resource Type	A named definition that fixes an Allocation Class and attributes.	name (unique); Allocation Class (Matched, Exclusive, Metered); attribute definitions (name and type: text, number, date, list); Unit Price (credits per instance-hour or unit-hour; none for Matched); bump cutoff (duration; Exclusive only); Refund Schedule (list of steps: minimum lead time, fraction $< 100\%$); retired
Resource	One thing that can be allocated.	name (unique within type); Resource Type; attribute values; Resource Owner (Member); units (integer, Metered only); retired from (date or none)
Pool Capacity	Derived: sum of units of a Metered type's unretired Resources.	integer, per Resource Type
Price Band	A time window carrying a multiplier on the Unit Price.	Resource Type; weekdays (subset of Mon–Sun); start and end time of day; multiplier (decimal > 0)
Quota	Optional per-Team cap on concurrent units of a Metered type.	Team; Resource Type; maximum concurrent units (integer)
Request	One request of a Team, submitted by its Manager.	Team; state (Draft, Submitted, Pending, Allocated, Waiting, Withdrawn, Refused); created at; submitted at; Objection Window end (timestamp or none); lines (one or more Request Lines); state history (list of: timestamp, actor, from state, to state)

Data dictionary (continued)

Element	Meaning	Composition and type
Request Line	One need inside a Request.	Resource Type; Allocation Class (derived); quantity (integer ≥ 1; one for a named Person); window (start and end timestamp); recurrence (weekday, time of day, first and last week; or none); Required or Optional; Priority line (yes or no; Exclusive only); skills (set) or named Person (Matched only); specific Resource (Exclusive, optional); locked Quote (credits or none until submitted); allocated instance (Resource or none); state (see below)
Line state	The state of one Request Line.	Open (in a Draft); Awaiting approval (Matched, approvals outstanding); Approved (Matched, all approvals in, Request not yet Allocated); Held (Exclusive or Metered, on Hold); Waitlisted; Allocated; Dropped (Optional, could not be satisfied); Released (in whole); Bumped (displaced, now Waitlisted at head); Expired (waitlisted past its window start)
Suggestion	A single proposed line from a Member to their Manager.	Member; Manager; proposed line (same fields as a Request Line); note; state (Open, Accepted, Declined); accepted into (Request Line or none); decline reason; created at
Candidate	A Person on a Shortlist for a Matched line, and the approval asked of their Manager.	Request Line; Member; person-days lent in the current Period (integer, at Shortlist time); selected (yes or no); decision (none, Approved, Refused); decided by (Member, or Capacity Planner on escalation); decided at; reason
Hold	A tentative, uncharged allocation.	Request Line; Resource (Exclusive) or units (Metered); placed at; expires at (Objection Window end)
Loan	A Person allocated to another Team for a window.	Request Line; Member (Person); lending Team; borrowing Team; start; end (may be shortened); person-days (derived)
Waitlist entry	A Required line waiting for a Resource Type and window.	Request Line; Resource Type; window; position (derived from Priority, Bump, submission time); joined at

Data dictionary (continued)

Element	Meaning	Composition and type
Account	A Team's standing in the Ledger.	Team; granted (credits); earned (credits); held (credits, derived from Holds); Balance and Available Balance (derived)
Transaction	One append-only entry against an Account.	Team; kind (Grant, Charge, Earning, Refund); amount (credits; negative only for an Earning correction); drawn from granted and from earned (Charge and Refund); Period; posted at; actor or job; Request Line (where applicable); corrects (Transaction or none); reason (Grant adjustment)
Period	The interval over which Grants are issued and expire.	start; end; Grants set per Team (credits); Day Rate in force (credits per person-day)
Statement	Derived per Team and Period from Transactions.	opening granted and earned; Grants; Charges; Earnings; Refunds; expired; closing granted and earned
Setting change	Audit record of a configuration change.	setting; old value; new value; actor; timestamp
Notification	An in-app message to a Member.	recipient; text; link to Request, line, or Suggestion; created at; read at

C Analysis Models

C.1 Request State Diagram

Figure 3 shows the states of a Request and every transition between them; Section 4.8 gives the guards in full. Three properties of the diagram carry the design: a Hold exists only while the Request is Pending, so no Hold outlives the Objection Window; Charges and Earnings post only on the transitions into Allocated; and Waiting is the single state from which a Request recovers, whether it lost a line to a refusal, a lapse, or a Bump.

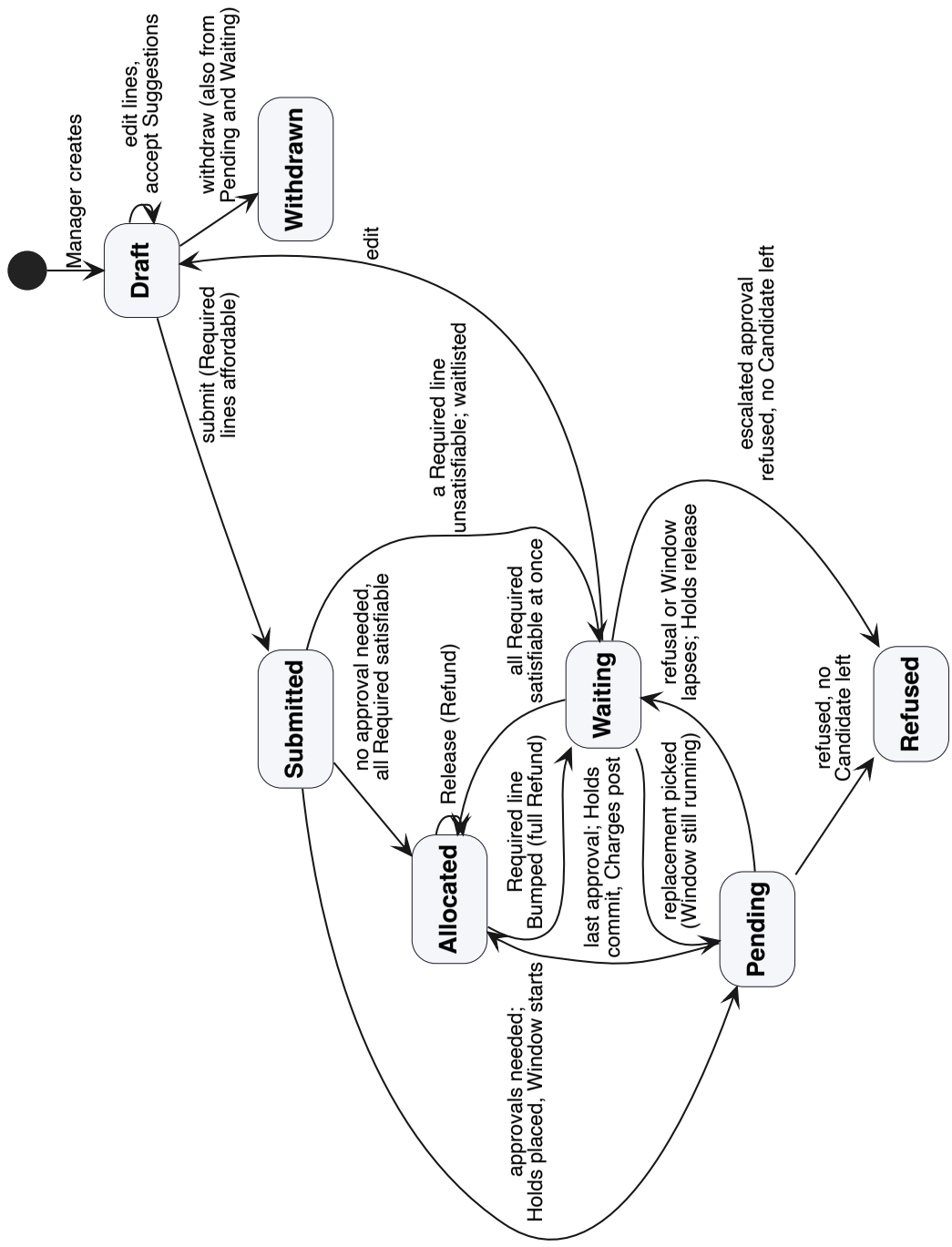


Figure 3: Request state-transition diagram. Submitted is a transient evaluation state left in the same transaction as it is entered. Withdraw is drawn once, from Draft; a Manager may also withdraw from Pending and Waiting. Allocated, Withdrawn, and Refused are final, except that Allocated returns to Waiting when a Required line is Bumped.

A typical Request with all three classes runs Draft → Submitted → Pending (room and GPUs on Hold, engineers awaiting approval) → Allocated (last approval arrives; Holds commit; Charges and Earnings post). The two failure paths are: a refusal, after which the Manager picks a replacement and the Request returns to Pending if the Objection Window is still running; and a lapse, after which the Capacity Planner decides the outstanding approvals and the Request is Allocated only when every Required line can be satisfied at once, with no Hold in between.

C.2 Use Case Diagram

Figure 4 shows the twelve use cases of Appendix D and the actors that start them. The Clock actor stands for the scheduled jobs that fire when an Objection Window lapses or a Hold expires; it has no user interface.

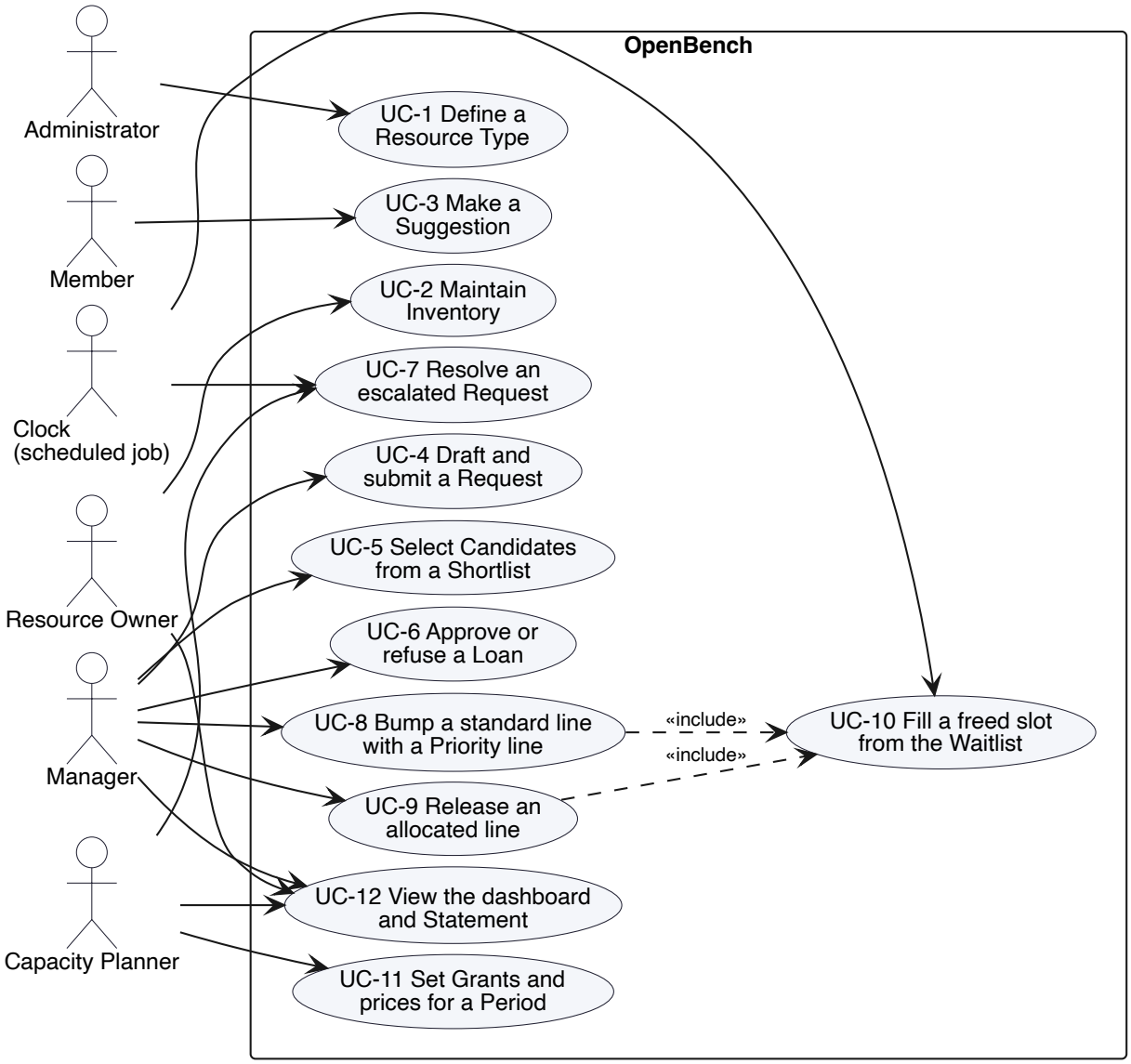


Figure 4: Use case diagram. All human actors are Members; the roles are additive.

C.3 Context Diagram and Data Model

The context diagram is Figure 1 in Section 3; the data model is Figure 2 in Appendix B.

D Use Cases

Each use case records its actors, trigger, preconditions, postconditions, flows, priority, frequency of use, business rules, and notes. Flow steps are numbered $n.0$ for the normal flow, $n.m$ for alternative flows, and $n.m$.E.k for exceptions. All use cases were created by the team on 2026-09-19 for version 0.1.

UC-1: Define a Resource Type

Use Case ID	UC-1
Use Case Name	Define a Resource Type
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Administrator (primary)
Description	An Administrator defines a new kind of Resource as data: its name, its Allocation Class, and its attributes. From then on Resource Owners can add Resources of that type and Managers can name it on a Request Line. No code changes.
Trigger	A new kind of room, instrument, or pool is to be offered through the system.
Preconditions	1. The Administrator is signed in. 2. No Resource Type of the same name exists.
Postconditions	1. The Resource Type is stored with its Allocation Class and attributes. 2. It appears in the Resource Type choice of the Inventory and Draft Request screens. 3. The change is recorded with actor and timestamp.
Normal Flow	**1.0 Define a type** 1. Administrator opens Administration, Resource Types, and chooses New. 2. System asks for a name, an Allocation Class, and attributes. 3. Administrator enters “Wet lab bench”, chooses Exclusive, and adds attributes *building* (text) and *fume hood* (list: yes, no). 4. Administrator saves. 5. System checks the name is unique, stores the type, and confirms. 6. System tells the Administrator that the Capacity Planner must set a Unit Price before lines of this type can be priced.
Alternative Flows	**1.1 Retire a type** 1. Administrator opens an existing type and chooses Retire. 2. System marks it retired; new lines cannot name it; existing allocations are unaffected.

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UC-1 Define a Resource Type (continued)

Exceptions	**1.0.E.1 Duplicate name** 1. System refuses to save and names the existing type. **1.1.E.1 Delete attempted** 1. The Administrator tries to delete a type referenced by a line; system refuses and offers Retire instead.
Includes	None
Priority	High
Frequency of Use	A few times per Period.
Business Rules	BR-4, BR-32
Special Requirements	Inventory.Type.NoCode: no deployment is needed.
Assumptions	The Administrator knows which Allocation Class fits; the field help explains the three.
Notes and Issues	The built-in type Person cannot be edited or retired.

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UC-2 Maintain Inventory (continued)

UC-2: Maintain Inventory

Use Case ID	UC-2
Use Case Name	Maintain Inventory
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Resource Owner (primary)
Description	A Resource Owner adds an Exclusive or Metered Resource, records the units a Metered Resource supplies, and sees who holds each Resource. The Resource Owner never approves an allocation.
Trigger	A Resource is put into service, changes size, or is taken out of service.
Preconditions	1. The Resource Owner is signed in. 2. The Resource Type exists.
Postconditions	1. The Resource is in the Inventory and is free for any window without allocations or Holds. 2. For a Metered Resource, the Pool Capacity of its type includes its units.
Normal Flow	**2.0 Add a Resource** 1. Resource Owner opens Inventory and chooses Add Resource. 2. System asks for a type, a name, and the type's attributes; for a Metered type it asks for units. 3. Resource Owner enters "GPU cluster B", type "GPU", units 16. 4. System stores the Resource with the Resource Owner as its owner and shows the type's new Pool Capacity.
Alternative Flows	**2.1 Change units** 1. Resource Owner opens the Resource and changes units from 16 to 8. 2. System checks every future hour; if the type's allocated plus held units never exceed the new Pool Capacity it saves the change. **2.2 View holders** 1. Resource Owner opens a Resource's timeline for a window. 2. System shows free, Hold, and allocated periods with the holding Team and Request.
Exceptions	**2.1.E.1 Reduction below use** 1. System refuses the change and reports the earliest hour in which allocated plus held units would exceed the new Pool Capacity, and by how much. **2.0.E.1 Retire with future use** 1. Resource Owner retires a Resource from a date with allocations or Holds after it; system refuses and names the blocking Request Lines.
Includes	None
Priority	High
Frequency of Use	A few times per Period per Resource Owner.
Business Rules	BR-4, BR-20, BR-32

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UC-2 Maintain Inventory (continued)

Special Requirements	None
Assumptions	Resource Owners keep the Inventory current (AS-2).
Notes and Issues	None

UC-3: Make a Suggestion

Use Case ID	UC-3
Use Case Name	Make a Suggestion
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Member (primary); Manager (secondary)
Description	A Member proposes a single Request Line to their own Manager. The Manager later accepts it into a Draft Request, possibly together with other Suggestions for the same thing, or declines it. Until accepted and submitted it has no effect on allocation.
Trigger	A Member sees a need for people, a room, or compute.
Preconditions	1. The Member is signed in and belongs to a Team with a Manager.
Postconditions	1. The Suggestion is stored as Open and listed for the Manager. 2. Nothing in the Inventory or Ledger has changed.
Normal Flow	**3.0 Suggest a line** 1. Member opens Suggestions and chooses New. 2. System asks for a Resource Type, quantity, window, and note; for the Person type it asks for skills or a named Person. 3. Member enters “GPU, 4 units, weeks 42–43, for the model sweep”. 4. Member submits; system stores the Suggestion as Open and notifies the Manager.
Alternative Flows	**3.1 Manager accepts** 1. Manager opens a Draft Request and the Open Suggestions panel. 2. Manager accepts the Suggestion as a new line, or into an existing line of the same type. 3. System marks it Accepted, links it to the line, keeps the original ask, and notifies the Member. **3.2 Manager declines** 1. Manager chooses Decline and optionally gives a reason. 2. System marks it Declined and notifies the Member.
Exceptions	**3.0.E.1 Invalid line** 1. The window is in the past or the quantity is zero; system refuses and explains.
Includes	None
Priority	Medium
Frequency of Use	A few per Member per Period.
Business Rules	BR-1

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UC-3 Make a Suggestion (continued)

Special Requirements	None
Assumptions	None
Notes and Issues	Declined and never-accepted Suggestions are the raw data for the Future unmet-demand analytics and are never deleted.

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UC-4 Draft and Submit a Request (continued)

UC-4: Draft and Submit a Request

Use Case ID	UC-4
Use Case Name	Draft and Submit a Request
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Manager (primary)
Description	A Manager assembles a Request of Required and Optional lines that may span all three Allocation Classes, sees a Quote per line against the Team's Available Balance, and submits. The system locks the Quotes, places Holds on the satisfiable Exclusive and Metered lines, and starts the approvals for the Matched lines.
Trigger	The Manager needs capacity the Team does not hold.
Preconditions	1. The Manager is signed in as the Manager of a Team. 2. The Capacity Planner has set Unit Prices for the Resource Types involved and the Team has an Account.
Postconditions	1. The Request is Pending, Allocated, or Waiting; every line's Quote is locked. 2. If Pending, every satisfiable Exclusive and Metered line is on Hold, the held credits equal their Quotes, the Objection Window has started, and lending Managers or the requesting Manager have been asked to act. 3. The submission is recorded with actor and timestamp.
Normal Flow	**4.0 Three-line Request** 1. Manager opens Requests and chooses New; system creates a Draft. 2. Manager adds a Matched line: skills PyTorch and Python, quantity 2, weeks 40–45. 3. Manager adds a Metered line: GPU, 4 units, weeks 42–43; system shows its Quote. 4. Manager adds an Exclusive line: Lab room, Thursdays 14:00–17:00, weeks 40–45; system shows its Quote. 5. Manager marks the room Optional; the other lines stay Required. 6. System shows the Required total, the Optional total, and the Available Balance; the Required total fits. 7. Manager submits. 8. System locks the Quotes, records the submission time, and evaluates every line. 9. The GPU line and the room are satisfiable; system places Holds on both and adds their Quotes to held credits. 10. The Matched line needs approval; system moves the Request to Pending, starts the Objection Window, and asks the Manager to select Candidates (UC-5).

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UC-4 Draft and Submit a Request (continued)

Alternative Flows	**4.1 Accept Suggestions** 1. At step 2 the Manager accepts Open Suggestions into new or existing lines (UC-3). **4.2 Split a line** 1. The Manager splits "GPU, 4 units" into "GPU, 2 units" Required and "GPU, 2 units" Optional. **4.3 Priority line** 1. The Manager marks the room a Priority line; its Quote includes the premium (UC-8). **4.4 No approval needed** 1. The Request has no Matched line and every Required line is satisfiable; system allocates it at once, posts the Charges, and shows it Allocated. **4.5 Required line unsatisfiable, no approval needed** 1. The Request has no Matched line and a Required room is taken; system waitlists the room, places no Holds, and shows the Request Waiting with the line's position. **4.6 Withdraw** 1. The Manager withdraws a Draft, Pending, or Waiting Request; system releases Holds and held credits and marks it Withdrawn. **4.7 Edit a Waiting Request** 1. The Manager edits a Waiting Request; system returns it to Draft, removes its lines from waitlists, and recomputes Quotes on resubmission.
Exceptions	**4.0.E.1 Unaffordable** 1. The Required total exceeds the Available Balance; system refuses submission, shows the shortfall, and the Request stays in Draft. **4.0.E.2 Invalid line** 1. A line's window starts in the past or its recurrence has no occurrence; system refuses to save it and explains.
Includes	UC-5
Priority	High
Frequency of Use	About 10,000 Requests per year: roughly four per Team per Period.
Business Rules	BR-1, BR-2, BR-3, BR-5, BR-6, BR-22, BR-24, BR-28
Special Requirements	QA-5: an untrained Manager completes flow 4.0 in under five minutes.
Assumptions	Grants have been posted for the current Period (AS-1).
Notes and Issues	Optional lines are never waitlisted; a Manager who wants to wait for something marks it Required.

UC-5: Select Candidates from a Shortlist

Use Case ID	UC-5
Use Case Name	Select Candidates from a Shortlist
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Manager (requesting, primary)
Description	For a skill-based Matched line the system shows the Candidates who hold every listed skill, are available for the whole window, and have no Loan overlapping it, ordered by fewest person-days lent in the current Period, and the requesting Manager picks the line's quantity of them. Each pick becomes an approval request to that Candidate's Manager.
Trigger	A Request with a skill-based Matched line enters Pending, or a refusal calls for a replacement.
Preconditions	1. The Request is Pending or Waiting and the Manager is signed in.
Postconditions	1. The selected Candidates are recorded; one approval request has been sent per Candidate to their Manager; each Candidate has been notified.
Normal Flow	**5.0 Select** 1. Manager opens the Matched line; system shows the Shortlist in order with Team, skills, and person-days lent in the current Period. 2. Manager selects two Candidates. 3. System records the selection, sends one approval request to each Candidate's Manager with the window and the Earning, and notifies the Candidates.
Alternative Flows	**5.1 Named Person** 1. The line names a Person; system skips the Shortlist, checks availability and that the Person has no overlapping Loan, and sends the approval request on entering Pending. **5.2 Replacement after refusal** 1. An approval was refused; system asks the Manager to select a replacement from the Shortlist; the Objection Window keeps running; if it is still running the Request returns to Pending with Holds re-placed.
Exceptions	**5.0.E.1 Too few Candidates** 1. The Shortlist holds fewer Candidates than the quantity; the line is unsatisfied; a Required line puts the Request in Waiting; system notifies the Manager when a new Candidate becomes available; the Manager may edit the line, which returns the Request to Draft. **5.0.E.2 Candidate no longer available** 1. Between Shortlist and selection the Candidate's availability changed or a Loan overlapping the window formed for them; system rebuilds the Shortlist and asks the Manager to select again.
Includes	UC-6
Priority	High
Frequency of Use	Once per skill-based Matched line, about half of all Requests.
Business Rules	BR-13, BR-12, BR-5, BR-8

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UC-5 Select Candidates from a Shortlist (continued)

Special Requirements	PE-3: the Shortlist appears within 3 s against 500 Members.
Assumptions	Members keep their skills and availability current.
Notes and Issues	Automatic selection was rejected in design: it treats people as interchangeable and removes the one point where the requesting Manager exercises judgment.

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UC-6 Approve or Refuse a Loan (continued)

UC-6: Approve or Refuse a Loan

Use Case ID	UC-6
Use Case Name	Approve or Refuse a Loan
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Manager (lending, primary); Manager (requesting) and Person (secondary)
Description	The Manager of a selected Candidate approves or refuses lending that Member for the window. Approval of the last outstanding Required approval allocates the Request: Holds commit, Charges post to the requesting Team, Earnings post to the lending Teams, and the Loans form. A refusal releases every Hold and sends the requesting Manager back to the Shortlist.
Trigger	An approval request arrives in the lending Manager's Approvals list.
Preconditions	1. The Request is Pending and its Objection Window is running. 2. The signed-in Manager leads the Candidate's Team, which is not the requesting Team.
Postconditions	1. The decision is recorded with actor, timestamp, and reason. 2. On approval of the last Required approval: the Request is Allocated, the Loans exist, and the Charges and Earnings are posted in one transaction. 3. On refusal: no Hold remains, held credits are zero for this Request, and the Request is Waiting.
Normal Flow	**6.0 Approve** 1. Lending Manager opens Approvals; system lists the request with requesting Team, Person, window, Earning, and time left. 2. Lending Manager chooses Approve. 3. System records the approval and marks the Candidate Approved. 4. This was the last Required approval; system commits the Holds, posts the Charges at the locked Quotes, forms the Loans, posts the Earnings, drops any Optional line that cannot be satisfied, moves the Request to Allocated, and notifies the requesting Manager, the lending Managers, and the Persons.
Alternative Flows	**6.1 Refuse** 1. Lending Manager chooses Refuse and gives a reason. 2. System releases every Hold and its held credits, moves the Request to Waiting, and asks the requesting Manager to select a replacement (UC-5 5.2). **6.2 Not the last approval** 1. Other Required approvals are outstanding; the Request stays Pending with its Holds. **6.3 Optional line approved after allocation** 1. The Request is already Allocated; system forms the Loan and posts the Earning now.

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UC-6 Approve or Refuse a Loan (continued)

Exceptions	**6.0.E.1 Objection Window lapsed** 1. The approval arrives after the lapse; system shows the request as escalated and routes the decision to the Capacity Planner (UC-7). **6.0.E.2 Own Team** 1. The requesting Manager is also the Candidate's Manager; system never sends the request and refuses the approval (BR-12).
Includes	None
Priority	High
Frequency of Use	One per selected Candidate.
Business Rules	BR-12, BR-14, BR-11, BR-10, BR-7, BR-6
Special Requirements	QA-2: step 4 is one transaction.
Assumptions	None
Notes and Issues	The Person is notified and has no action; consent is a Future item.

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UC-7 Resolve an Escalated Request (continued)

UC-7: Resolve an Escalated Request

Use Case ID	UC-7
Use Case Name	Resolve an Escalated Request
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Capacity Planner (primary); Clock (trigger); both Managers (secondary)
Description	When the Objection Window lapses with a Required approval outstanding, the system releases every Hold, moves the Request to Waiting, and escalates the outstanding approvals to the Capacity Planner, who approves or refuses on the silent Manager's behalf. The Request is then allocated only at a moment when every Required line can be satisfied at once.
Trigger	The scheduled job finds a Pending Request whose Objection Window end has passed.
Preconditions	1. The Request is Pending with at least one Required approval outstanding.
Postconditions	1. No Hold remains for the Request; held credits for it are zero. 2. The Request is Waiting, Allocated, or Refused, and every decision is recorded with the Capacity Planner as actor.
Normal Flow	**7.0 Escalate and approve** 1. Within five minutes of the lapse the system releases the Holds, moves the Request to Waiting, and notifies the requesting Manager, the silent Manager, and the Capacity Planner. 2. Capacity Planner opens the escalations list; system shows the Request, the outstanding approvals, the silent Manager, and the time since the lapse. 3. Capacity Planner approves the outstanding approval. 4. System re-evaluates the Request: every Required line can be satisfied now (the room and GPUs are still free); system allocates it at the locked Quotes, posts Charges and Earnings, forms the Loans, and notifies everyone.
Alternative Flows	**7.1 Capacity no longer free** 1. At step 4 the room has been taken; system waitlists the room line; the Request stays Waiting and is allocated when the waitlist fills the room and every other Required line still fits (UC-10). **7.2 Capacity Planner refuses** 1. System asks the requesting Manager for a replacement; further approvals on this Request go to the Capacity Planner directly; if no Candidate remains the Request becomes Refused. **7.3 Settle with a Grant** 1. The Capacity Planner posts a mid-Period Grant to one of the Teams with a reason, for example to compensate a Team whose Hold was lost.

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UC-7 Resolve an Escalated Request (continued)

Exceptions	**7.0.E.1 Silent Manager decides after the lapse** 1. System shows the silent Manager that the decision has moved to the Capacity Planner and accepts no decision from them.
Includes	None
Priority	Medium
Frequency of Use	Rare; the Objection Window is meant to make it rare.
Business Rules	BR-8, BR-6, BR-7, BR-32
Special Requirements	PE-4: step 1 happens within five minutes of the lapse.
Assumptions	None
Notes and Issues	After the lapse no Hold is placed again for this Request; this is what keeps every Hold bounded.

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UC-8 Bump a Standard Line with a Priority Line (continued)

UC-8: Bump a Standard Line with a Priority Line

Use Case ID	UC-8
Use Case Name	Bump a Standard Line with a Priority Line
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Manager (requesting, primary); Manager (Bumped, secondary)
Description	A Manager marks an Exclusive line as a Priority line and carries the premium. If no instance is free, the system Bumps the standard line that blocks it, outside the bump cutoff: the Bumped line is refunded in full and goes to the head of the waitlist, and its Manager is notified. Priority lines cannot be Bumped.
Trigger	A Priority line is evaluated and finds no free instance.
Preconditions	1. The Team can afford the Priority line's Quote, premium included. 2. The blocking line is a standard line whose window starts later than the bump cutoff from now.
Postconditions	1. The Priority line is on Hold or allocated on the freed instance. 2. The Bumped line is at the head of its waitlist, its Charge is refunded in full, and its Request is Waiting if the line was Required in an Allocated Request. 3. The Bump is recorded with both lines, actor, and timestamp.
Normal Flow	**8.0 Bump** 1. Manager marks the Lab room line as a Priority line in the Draft; system shows the Quote with the premium. 2. Manager submits (UC-4); the Team can afford it. 3. System finds no free instance for the window. 4. System chooses the instance that needs the fewest Bumps, preferring the latest-submitted standard lines, and checks each is outside the bump cutoff. 5. System Bumps the standard line: posts a full Refund, places it at the head of the waitlist, notifies its Manager with the reason, and returns its Request to Waiting if the line was Required. 6. System places the Priority line on Hold or allocates it as the rest of the Request requires.
Alternative Flows	**8.1 Inside the cutoff** 1. The blocking line starts within the bump cutoff; system does not Bump; the Priority line joins the waitlist ahead of every standard line. **8.2 Blocked by a Priority line** 1. The blocking line is itself a Priority line; no Bump; the new line waits behind it. **8.3 Bumped line finds another instance** 1. The waitlist offers the Bumped line another free instance at once (UC-10); it is satisfied there and its Request re-evaluated.

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UC-8 Bump a Standard Line with a Priority Line (continued)

Exceptions	**8.0.E.1 Unaffordable** 1. The premium Quote exceeds the Available Balance; submission is refused (UC-4 4.0.E.1).
Includes	UC-4, UC-10
Priority	Medium
Frequency of Use	Rare; the premium is meant to make it rare.
Business Rules	BR-15, BR-16, BR-17, BR-18, BR-19
Special Requirements	Step 5 is one transaction.
Assumptions	None
Notes and Issues	The Bumped Team is not penalized: full Refund, head of the waitlist. The disruption of being Bumped, rather than any deduction of credits, is what discourages gambling on a standard line for critical work.

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UC-9 Release an Allocated Line (continued)

UC-9: Release an Allocated Line

Use Case ID	UC-9
Use Case Name	Release an Allocated Line
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Manager (primary)
Description	A Manager gives back an allocated Exclusive or Metered line, in whole or in part, before it is consumed, and receives a partial Refund on the Resource Type's Refund Schedule. A Manager may also end a Loan early, which prorates the lending Team's Earning.
Trigger	The Team no longer needs some or all of what it was allocated.
Preconditions	1. The line is Allocated and the part to release has not started.
Postconditions	1. The released capacity is free and has been offered to the waitlist. 2. A Refund is posted to the part of the Account the Charge was drawn from. 3. The Release is recorded with lead time, fraction, and amount.
Normal Flow	**9.0 Release a whole line** 1. Manager opens the Allocated Request and chooses Release on the GPU line, ten days before it starts. 2. System shows the lead time (10 days), the fraction from the Refund Schedule (75%), and the Refund amount. 3. Manager confirms. 4. System posts the Refund to granted and earned in the proportions of the Charge, marks the line Released, frees the units, and offers them to the waitlist (UC-10).
Alternative Flows	**9.1 Partial Release** 1. Manager shortens the window, removes occurrences, or reduces units; system computes the Refund on the released part only; the rest keeps its allocation and prices. **9.2 End a Loan early** 1. Manager ends a Loan from a chosen date; system shortens the Loan, posts an Earning correction to the lending Team for the days no longer lent, and notifies the lending Manager.
Exceptions	**9.0.E.1 Already started** 1. The window has started; the fraction is 0%; system still frees the remaining capacity if the Manager confirms, and posts no Refund. **9.0.E.2 Window ended** 1. Nothing remains to release; system refuses.
Includes	UC-10
Priority	High
Frequency of Use	A fraction of allocations per Period.
Business Rules	BR-25, BR-26, BR-11, BR-29

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UC-9 Release an Allocated Line (continued)

Special Requirements	None
Assumptions	None
Notes and Issues	Capacity that is neither used nor Released is never freed automatically; it shows as idle on the dashboard.

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UC-10 Fill a Freed Slot from the Waitlist (continued)

UC-10: Fill a Freed Slot from the Waitlist

Use Case ID	UC-10
Use Case Name	Fill a Freed Slot from the Waitlist
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Clock and system events (primary); Manager of the waiting line (secondary)
Description	When capacity frees through a Release, a Bump, or an expired Hold, the system offers it to the waiting lines for that Resource Type in order. The first line whose window fits is satisfied: if that completes its Request, the Request is allocated at its locked Quotes; if its Request is still awaiting approvals inside the Objection Window, the line is placed on Hold; otherwise the slot passes to the next line and positions are kept.
Trigger	A Release, a Bump, a Hold expiry, or a Withdrawal frees capacity for which a waitlist exists.
Preconditions	1. At least one line is waiting for the Resource Type with a window overlapping the freed capacity.
Postconditions	1. The freed capacity is allocated, held, or remains free; no waiting line has lost its position without being satisfied or expiring. 2. Every affected Manager has been notified.
Normal Flow	**10.0 Fill and allocate** 1. Capacity frees. 2. System takes the waitlist for the Resource Type in order: Priority lines, then submission time, Bumped lines at the head. 3. The first line whose window fits belongs to a Waiting Request whose other Required lines are all satisfied. 4. System allocates the line and the Request, posts the Charges at the locked Quotes, and notifies the Manager.
Alternative Flows	**10.1 Request still awaiting approvals** 1. The line's Request is inside its Objection Window with approvals outstanding; system places the line on Hold and moves the Request to Pending. **10.2 Request cannot complete** 1. The Request has other unsatisfied Required lines and no running Objection Window; system offers the capacity to the next waiting line; the skipped line keeps its position. **10.3 Expiry** 1. A waiting line's window start has passed; system removes it, marks it Expired, notifies the Manager; the Request stays Waiting.
Exceptions	**10.0.E.1 Team can no longer afford it** 1. The Team's Available Balance is below the locked Quotes of the Required lines; system skips the line, keeps its position, and notifies the Manager.
Includes	None

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UC-10 Fill a Freed Slot from the Waitlist (continued)

Priority	High
Frequency of Use	Once per freed slot with a non-empty waitlist.
Business Rules	BR-18, BR-6, BR-7, BR-24, BR-28
Special Requirements	PE-4 for Hold expiry; a Release or Bump triggers the fill in the same transaction.
Assumptions	None
Notes and Issues	Because a skipped line keeps its position, a Request waiting for two things is allocated at the first moment both are free at once.

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UC-11 Set Grants and Prices for a Period (continued)

UC-11: Set Grants and Prices for a Period

Use Case ID	UC-11
Use Case Name	Set Grants and Prices for a Period
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Capacity Planner (primary); Clock (rollover)
Description	The Capacity Planner sets the levers that shape demand: the Period, each Team's Grant, Unit Prices, Price Bands, the Day Rate, Refund Schedules, Quotas, the priority premium, bump cutoffs, and the Objection Window. At the Period boundary the system expires unspent granted credits and posts the new Grants.
Trigger	A new Period approaches, or demand patterns call for a change.
Preconditions	1. The Capacity Planner is signed in.
Postconditions	1. Settings are stored with their change history. 2. At rollover every Team's granted credits are zeroed, the expired amount is on the Statement, the new Grant is posted, and earned credits are unchanged.
Normal Flow	**11.0 Prepare a Period** 1. Capacity Planner opens Settings. 2. Capacity Planner enters each Team's Grant for the coming Period. 3. Capacity Planner sets or reviews Unit Prices and Price Bands per Resource Type, the Day Rate, Refund Schedules, Quotas, the premium, cutoffs, and the Objection Window. 4. System validates: bands do not overlap, no Refund step reaches 100%, values are positive. 5. Capacity Planner saves; system records each change. 6. At the Period boundary the system, within fifteen minutes, expires granted credits, posts the Grants, and opens the new Statements.
Alternative Flows	**11.1 Mid-Period Grant** 1. Capacity Planner posts an additional Grant to one Team with a reason; it is added to granted credits and expires with them.
Exceptions	**11.0.E.1 Invalid setting** 1. Overlapping bands or a Refund step of 100% or more; system refuses and names the field. **11.0.E.2 Missing Grant** 1. A Team has no Grant set when the Period starts; system posts a Grant of zero and notifies the Capacity Planner.
Includes	None
Priority	High
Frequency of Use	Once per Period plus occasional adjustments.
Business Rules	BR-31, BR-27, BR-23, BR-25, BR-15, BR-16, BR-8, BR-21, BR-32
Special Requirements	PE-5

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UC-11 Set Grants and Prices for a Period (continued)

Assumptions	None
Notes and Issues	Price changes never alter a locked Quote.

UC-12: View the Dashboard and Statement

Use Case ID	UC-12
Use Case Name	View the Dashboard and Statement
Created By	J. Nana, B. Lujan, F. Turopov, Y. Shnazbayev, 2026-09-19
Actors	Manager, Capacity Planner, Resource Owner (primary)
Description	A viewer sees utilization and idle capacity per Resource Type, credits earned and spent per Team, and the per-Period Statement with every Transaction behind it. What is visible depends on the viewer's role.
Trigger	The viewer opens the Dashboard or Statement screen.
Preconditions	1. The viewer is signed in with one of the three roles.
Postconditions	1. None; the use case changes no data.
Normal Flow	**12.0 Manager's view** 1. Manager opens the Dashboard; system shows the current Period's utilization and idle capacity for every Resource Type, the Team's Grants, Charges, Earnings, Refunds, Balance and Available Balance, and person-days lent and borrowed. 2. Manager opens the Statement; system shows the Period's opening amounts, every Transaction, the expired amount, and the closing amounts. 3. Manager selects a past Period; system shows the same for that Period.
Alternative Flows	**12.1 Capacity Planner** 1. System shows every Team's figures and Statements, and the waitlist lengths per Resource Type. **12.2 Resource Owner** 1. System shows utilization and idle capacity for the Resources the viewer owns.
Exceptions	**12.0.E.1 Another Team's Statement** 1. A Manager requests another Team's Statement through the API; system refuses (SE-5).
Includes	None
Priority	Medium
Frequency of Use	Weekly per Manager; daily for the Capacity Planner.
Business Rules	BR-30, BR-27, BR-32
Special Requirements	PE-2
Assumptions	None
Notes and Issues	Requested-versus-used figures appear only with the Stretch usage recording.

E Inspection Report

This appendix records the findings of the team's inspection of this document and their resolution. It is filled in by the inspectors, not generated. One row per finding.

Categories: ambiguity (more than one reading), incompleteness (a case or requirement missing), incorrectness (does not describe the intended behavior), inconsistency (contradicts another statement or the glossary), infeasibility (cannot be built in the semester), traceability (a requirement with no feature, use case, or rule behind it), clarity (correct but hard to read). **Status:** Open, Closed (with the version in which it was fixed), or Rejected (with the reason in the Resolution column).

All four team members, J. Nana, B. Lujan, F. Turopov, and Y. Shnazbayev, each read the whole document and took part in the inspection meeting on 2026-09-21, at which the findings were categorized and resolved.

ID	Found by	Location	Category	Finding	Resolution	Status

Usability Test Record

QA-5 requires a test with at least three people from outside the team before the final demonstration. Record each session here: date, participant's role or background, time to draft and submit the three-line Request, and what they got stuck on.

Date	Participant	Time taken	Observations

F Traceability

F.1 Features to Requirements, Use Cases, and Business Rules

Each major feature of Section 1 maps to one feature section, whose functional requirements share an identifier prefix, to the use cases that exercise it, and to the business rules it depends on. A requirement outside this table is a defect to record in Appendix E.

Feature	Section	Requirement prefix	Use cases	Business rules
FE-1	4.1 Inventory and Resource Types	Inventory.*	UC-1, UC-2	BR-4, BR-20, BR-32
FE-2	4.2 Member Profiles	Profile.*	UC-5	BR-13
FE-3	4.3 Suggestions	Suggest.*	UC-3, UC-4	BR-1
FE-4	4.4 Draft and Submit a Request	Draft.*	UC-4, UC-8	BR-1, BR-2, BR-3, BR-5, BR-22, BR-24, BR-28
FE-5	4.5 Matched Allocation	Matched.*	UC-5, UC-6, UC-7	BR-5, BR-8, BR-10, BR-11, BR-12, BR-13, BR-14
FE-6	4.6 Exclusive Allocation	Exclusive.*	UC-4, UC-8, UC-10	BR-3, BR-15, BR-16, BR-17, BR-18, BR-19
FE-7	4.7 Metered Allocation	Metered.*	UC-4, UC-10	BR-3, BR-20, BR-21, BR-19
FE-8	4.8 Request Lifecycle and Holds	Lifecycle.*	UC-4, UC-6, UC-7, UC-10	BR-2, BR-6, BR-7, BR-8, BR-9, BR-17
FE-9	4.9 Release and Refund	Release.*	UC-9, UC-10	BR-25, BR-26, BR-11, BR-29
FE-10	4.10 Ledger and Pricing	Ledger.*	UC-11, UC-12, UC-6	BR-10, BR-11, BR-22, BR-23, BR-24, BR-26, BR-27, BR-28, BR-29, BR-30, BR-31
FE-11	4.11 Showback Dashboard	Dashboard.*	UC-12	BR-30, BR-32
FE-12	4.12 Administration	Admin.*, Planner.*	UC-1, UC-11, UC-7	BR-32, BR-31, BR-23, BR-25, BR-8

F.2 Business Objectives to Features

Objective	Achieved through	Success criteria
BO-1	FE-1, FE-6, FE-7, FE-9, FE-10 (Price Bands), FE-11	SC-2
BO-2	FE-2, FE-5, FE-10 (Day Rate, earned credits carry over)	SC-1
BO-3	FE-3, FE-4, FE-8, FE-12 (Objection Window, escalations)	SC-3, SC-4

F.3 Decisions Behind the Requirements

The following design decisions are recorded as architecture decision records in the project repository under `docs/adr/` and are the rationale for the rules named: Holds bounded by the Objection Window (BR-6, BR-8); the two-part Account with expiring Grants and carried Earnings (BR-27, BR-26); priority bought with a premium rather than granted by rank (BR-15); and Matched lines costing nothing (BR-10, BR-11).